

CONTINUOUSLY MONITORED BARRIER OPTIONS UNDER MARKOV PROCESSES

UNABRIDGED VERSION WITH MATLAB CODE

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ABSTRACT. In this paper we present a fast and accurate algorithm for pricing barrier options in one-dimensional Markov models, including general local volatility models with jumps, Lévy processes and Lévy driven SDEs. The approach rests on the construction of an approximating continuous-time Markov chain that closely follows the dynamics of the given Markov model. We illustrate the method by implementing it for a range of models, including a local Lévy process and a local volatility jump-diffusion. Code in Matlab for one of the numerical examples is included in the paper (and is also available online). We also provide a convergence proof and error estimates for this algorithm.

1. INTRODUCTION

1.1. Background and motivation. Barrier options are among the most popular exotic derivatives. Such contracts form effective risk management tools, and are liquidly traded in the Foreign Exchange markets. The most liquid barrier options in FX markets are continuously monitored single- or double-no-touch options and knock-in or knock-out calls and puts (see e.g. Hakala and Wystup [30], Lipton [48], [49], Wystup [60]). The main challenge in the risk management of large portfolios of barrier options faced by trading desks that make a market in these securities is to be able to price and hedge the barrier products in models that are flexible enough to describe the observed option prices (i.e. calibrate to the vanilla market price quotes).

It is by now well established that the classical Black-Scholes model lacks the flexibility to fit accurately to observed option price data (see e.g. Gatheral [25] and the references therein). A variety of models have been proposed to provide an improved description of the dynamics of the price of the underlying that can more accurately describe the option surface. Parametric diffusion models like the CEV process [16] have additional flexibility to fit the vanilla skew at a single maturity for as many options as there are free parameters in the model. The seminal idea (developed by Dupire [21] and Gyöngy [28]) that allows one to construct a model that can describe the entire implied volatility surface (across all strikes and maturities) is that of local volatility models, where a non-parametric form of the local volatility function is constructed from the option price data. It

Acknowledgements: We would like to thank the anonymous referees, Bjorn Eriksson, Mike Giles, Vassili Kolokoltsov, Steven Kou, Sergei Levendorskii, Dilip Madan, Jan Obloj and Johannes Stolz as well as the participants of the 2009 Leicester workshop on Spectral and Cubature Methods in Finance and Econometrics for useful suggestions and constructive comments, which led to improvements of the paper. Research supported by EPSRC grant EP/D039053. The research was conducted while the first author was at the Mathematics Department of Imperial College London.

has been shown that in practice such models imply unrealistic dynamics of the option prices (see the formula for the implied volatility in a local volatility model given in [29]). The ramification is an unrealistic amount of vega risk, which is expensive to hedge. Therefore, even though in a local volatility model barrier options can be priced using a PDE solver, this modelling framework alone is not suitable for the risk management of a large portfolio of barrier options.

At the other end of the spectrum are the jump processes with stationary and independent increments, which can fit very well the volatility smile at a single maturity (see e.g. [14] and the references therein). A variety of models in the exponential Lévy class have been proposed in the literature: CGMY [10], KoBoL [7], generalised hyperbolic [22], NIG [5] and Kou [39]. Exponential Lévy processes are simple examples of Markov processes whose law is uniquely determined by the distribution of the process at a single time. Since the set of call option prices at a fixed maturity for all strikes uniquely determines the marginal risk-neutral distribution at that maturity, calibration to option prices at multiple maturities in principle fixes the corresponding marginals. It has been reported (see e.g. [14]) that Lévy processes lack the flexibility of calibrating simultaneously across a range of strikes and maturities. Several generalisations within the one-dimensional Markov framework have been proposed.

If the stationarity assumption is relaxed while the property of independence of increments is retained, one arrives at the class of exponential additive processes, which have recently been shown to calibrate well to several maturities in equity markets. The Sato process introduced in Carr et al. [12] is an example of such an additive model used in financial modelling.

The independent increments property of a process implies that its transition probabilities are translation invariant in space, so that they only depend on the difference between the end and starting value of the process. It is well known that the distribution of a log-asset price depends in a non-linear way on the starting point (e.g. in equity markets it has been observed that if the current price is high, then the volatility is low and vice versa). To capture this effect one is led to consider Markov jump-processes whose increments are not independent. As a generalisation of local volatility models, the class of local Lévy processes introduced by Carr et al. [11] allows the modeller to modulate the intensity of the jumps as well as their distribution depending on where the underlying asset is trading. A local volatility jump-diffusion with similar structural properties was calibrated to the implied volatility surface in Andersen and Andreasen [3] and He et al. [31]. Due to the presence of jumps and the absence of stationarity and independence of increments, the problem of obtaining the first-passage probabilities for such a general class of processes is computationally less tractable.

There exists currently a good deal of literature on numerical methods for the pricing of barrier-type options. It is well known that in this case a straightforward Monte Carlo simulation algorithm will be time-consuming and yield unstable results for the prices and especially the sensitivities. The knock-in/out features in the barrier option payoffs lead to slower convergence of the Monte Carlo algorithm. To address this problem the following (semi-)analytical approaches have been developed for specific models:

- (a) spectral expansions for several parametric diffusion models (Davydov and Linetsky [18], Lipton [48]),
- (b) transform based approaches for exponential Lévy models (Boyarchenko and Levendorskii [6], Geman and Yor [26], Jeannin and Pistorius [37], Kou and Wang [40], Sepp [58]),

The method (a) employs the explicit spectral decompositions for this class of diffusion models, whereas the approach (b) exploits the independence and stationarity of the increments of the Lévy process, and the so-called Wiener-Hopf factorisation. Since both of these approaches hinge on special structural properties of the underlying processes, it is not clear if and how they can be extended to more general Markovian models.

A different approach, pioneered by Kushner (see e.g. [43]), is the discrete time Markov chain approximation method. Originally developed for the numerical solution of stochastic optimal control problems in continuous time, this method consists of approximating the system of interest by a discrete time chain that closely follows its dynamics, and solving the problem of interest for this chain. An application to the pricing of American type options is given in Kushner [42]. Bally et al. [4] develop a quantization method to efficiently value American options on baskets of assets under a local volatility model. Using a discrete time Markov chain, Duan et al. [19] price a discretely monitored barrier option in the Black-Scholes and NGARCH models. Rogers and Stapleton [55] investigate an efficient binomial tree method for barrier option pricing (see also references therein for related methods). Related are PDE and PIDE finite difference discretization methods that have been investigated by various authors; Zvan et al. [61] consider barrier and related options in the Black-Scholes model; Tavella and Randall [59] present an overview of PDE finite difference methods for the pricing of financial instruments; Wang et al. [34] develop a robust scheme for American options under a CGMY model, and Cont and Voltchkova [15] follow a viscosity approach for the PIDEs connected to European and barrier option under Lévy models. Markov chains have also been employed to directly model the evolution of price processes; Albanese and Mijatović [2] model the stochasticity of risk reversals and carried out a calibration study in FX markets under a certain continuous-time Markov chain constructed to model the FX spot process.

1.2. Contribution of the current paper. In this paper we consider the problem of pricing barrier options in the setting of one-dimensional Markov processes, which in particular includes the case of local Lévy models as well as local volatility jump-diffusions and additive processes. The presented approach is probabilistic in nature and is based on the following two elementary observations: (i) given a Markov asset price process S it is straightforward to construct a continuous-time Markov chain model X whose law is close to that of S , by approximating the generator of the process S with an intensity matrix; (ii) the corresponding first-passage problem for a continuous-time Markov chain can be solved explicitly via a closed-form formula that only involves the generator matrix of the chain X . More precisely, for a given Markov asset price model S on the state-space $\mathbb{E} = [0, \infty)$ with corresponding generator $\mathcal{L}$ the algorithm for the pricing of any barrier product (including rebate options, which depend on the position at the moment of first-passage) consists of the following two steps:

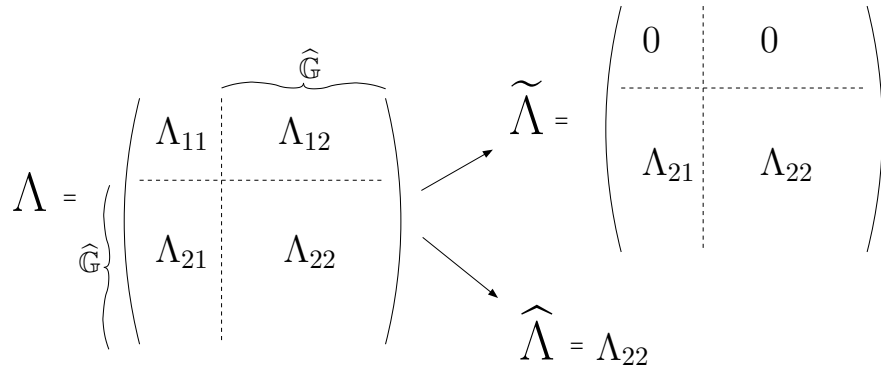


FIGURE 1. This is a schematic picture in bloc notation that demonstrates how the matrices $\hat{\Lambda}$ and $\tilde{\Lambda} = \tilde{\Lambda}_0$ are obtained from the generator Λ . The matrix Λ is the generator matrix of an approximating continuous-time Markov chain X on the state-space $\mathbb{G}$. The subset $\hat{\mathbb{G}} \subset \mathbb{G}$ consists of the elements of $\mathbb{G}$ that lie between the barriers. The matrix $\hat{\Lambda}$ contains all the information necessary to price any contract that knocks out (or knocks in) when a barrier is breached. Similarly the matrix $\tilde{\Lambda}$ is what is required to compute the distribution of any function that depends on the first-passage and overshoot of the chain X into the region on the other side of the barriers.

- (i) Construct a finite state-space $\mathbb{G} \subset \mathbb{E}$ and a generator matrix Λ for the chain X that approximates the operator $\mathcal{L}$ on $\mathbb{G}$.
- (ii) To value knock-out and rebate options, obtain the matrices $\hat{\Lambda}$ and $\tilde{\Lambda}$ by the procedure in Figure 1 and apply closed-form formulas in terms of these matrices (given in equations (3.9) and (3.11) below).

The form of the generators of Markov processes that commonly arise in pricing theory (including the local Lévy class) is well known from general theory (they are reviewed in Sections 2 and 4). The state-space $\mathbb{G}$ in step (i) is taken to be non-uniform with a higher density of points in the relevant areas such as spot and barrier levels. Subsequently, the generator matrix Λ is defined by matching the instantaneous moments of the Markov processes S and the chain X on the state-space $\mathbb{G}$. This criterion implies in particular that the chain X has the same average drift as the asset price process S . Step (ii) of the algorithm consists of the evaluation of the closed-form formulas for the first-passage probabilities that can be derived employing continuous-time Markov chain theory (see Theorem 1 below). The evaluation of this formula consists of exponentiation of either the matrix $\hat{\Lambda}$ or $\tilde{\Lambda}$, which can be performed using the Padé approximation algorithm that is implemented in standard packages such as Matlab (see also [32]). The outputs of the algorithm yield arbitrage-free prices in a certain continuous time Markov chain model for the risky asset price process. This feature is a consequence of the fact that the algorithm is based on the construction of an approximate stochastic model. This property is not shared by many other numerical methods used in practice that are based on purely analytical considerations.

We implemented this algorithm for a number of models that include the features of local volatility, jumps and time-dependent characteristics. We obtained an accurate match with the numerical results under diffusion and Lévy models that were considered elsewhere in the literature (see Section 7). We prove that by refining the grid the prices generated by this approach converge to those of the limiting model. We also establish, under additional regularity assumptions, an error bound

that is linear in the spatial mesh size and the truncation error, and in the mesh of the time-grid if the process is time-inhomogeneous (see Section 6). We showed that an additional logarithmic factor may arise in this error bound when the Lévy density has a pole of order two at the origin. Numerical experiments (reported in Section 7) appear to suggest that, for a number of models, the error actually decays quadratically in the spatial mesh size.

There is good deal of literature devoted to the study of the (weak) convergence of Markov chains to limiting processes. However the (sharp) rates of convergence of prices generated by the Markov chain approximation to those of the limiting model are rarely available, especially for barrier options. For discrete time Markov chains some explicit rates have been established (see e.g. Broadie et al. [8] and Gobet and Menozzi [27]). Establishing the sharp rates of convergence for specific models remains an open question, left for future research.

The remainder of the paper is organized as follows. In Section 2 we define the class of models and barrier option contracts that is considered, and state some preliminary results about Markov processes. Section 3 presents the formulas for the first-passage quantities of the continuous-time Markov chains. In Section 4 we extend the approach to time-inhomogeneous Markov processes. In Section 5 we describe the discretization algorithm to construct the intensity matrix Λ of the chain X . Section 6 states the convergence results, which are proved in Appendix A. Numerical results are presented in Section 7 and Section 8 concludes the paper.

2. PROBLEM SETTING: BARRIER OPTIONS FOR MARKOV PROCESSES

The problem under consideration is that of the valuation of general barrier options, which can be formulated as follows. Given a random process $S = \{S_t\}_{t \geq 0}$ modelling the price evolution of a risky asset, non-negative payoff and rebate functions g and h , and a set A specifying the range of values for which the contract ‘knocks out’, it is of interest to evaluate the expected discounted value of the random cash flow associated to a general barrier option contract

$$(2.1) \quad g(S_T)\mathbf{I}_{\{\tau_A > T\}} + h(S_{\tau_A})\mathbf{I}_{\{\tau_A \leq T\}},$$

where $\mathbf{I}_C$ denotes the indicator of a set C and

$$\tau_A = \inf\{t \geq 0 : S_t \in A\}$$

is the first time that S enters the set A . Furthermore, it is relevant to quantify the sensitivities of this value with respect to different parameters such as the spot value $S_0 = x$. The cash flow in (2.1) consists of a payment $g(S_T)$ in the case the contract has not knocked out by the time T , and a rebate $h(S_{\tau_A})$ if it has. Examples of commonly traded options included in this setting are the down-and-out, up-and-out and double knock-out options. In particular, by taking $A = \emptyset$ we retrieve the case of a standard European claim with payoff $g(S_T)$ at maturity T .

We will consider this valuation problem in a Markovian setting, assuming that the underlying S is a Markov process with state-space $\mathbb{E} := [0, \infty)$ defined on some filtered probability space $(\Omega, \mathcal{F}, \mathbf{F}, \mathbf{P})$, where $\mathbf{F} = \{\mathcal{F}_t\}_{t \geq 0}$ denotes the standard filtration generated by S . Thus, S takes

values in $\mathbb{E}$ and satisfies the Markov property:

$$(2.2) \quad \mathbf{E}[f(S_{t+s})|\mathcal{F}_t] = P_s f(S_t),$$

for all $s, t \geq 0$ and bounded Borel functions f , where $\mathbf{E}$ denotes the expectation under the probability measure $\mathbf{P}$ and $P_s f$ is given by

$$(2.3) \quad P_s f(x) := \mathbf{E}_x[f(S_s)] := \mathbf{E}[f(S_s)|S_0 = x].$$

By taking expectations in (2.2) we see that the family $(P_t)_{t \geq 0}$ forms a semigroup:

$$P_{t+s}f = P_t(P_s f), \quad \text{for all } s, t \geq 0, \text{ and } P_0 f = f.$$

Informally, these conditions state that the expected value of the random cash flow $f(S_{t+s})$ occurring at time $t + s$ conditional on the available information up to time t depends on the past via the value S_t only. Setting the rate of discounting equal to a non-negative constant r , for any pair of non-negative Borel functions g and h the expected discounted value of the barrier cash flow (2.1) at the epoch $\tau_A \wedge T$, the earlier of maturity T and the first entrance time τ_A , is given by

$$(2.4) \quad \mathbf{E}_x \left[e^{-rT} g(S_T) \mathbf{I}_{\{\tau_A > T\}} \right] + \mathbf{E}_x \left[e^{-r\tau_A} h(S_{\tau_A}) \mathbf{I}_{\{\tau_A \leq T\}} \right].$$

If S represents the price of a tradeable asset, r is the risk-free rate, d is the dividend yield and the process $\{e^{-(r-d)t} S_t\}_{t \geq 0}$ is a martingale, standard arbitrage arguments imply that no arbitrage is introduced if expression (2.4) is used as the current price of the option with payoff (2.1).

Before proceeding we review some key concepts of the standard Markovian setup that will be needed in the sequel. For background on the (general) theory of Markov processes we refer to the classical works Chung and Walsh [13], Ethier and Kurtz [23], Itô and McKean [35] and Rogers and Williams [56] (the latter two in particular treat the case of Markov processes with continuous sample paths). In what follows we will restrict S to be in a subclass of Markov processes for which, if the function f is continuous and tends to zero at infinity, the expected payoff $P_t f(x)$ has the following properties: it depends continuously on the spot $S_0 = x$ and on expiry t and also decays to zero when x tends to infinity. More precisely, denoting by $C_0(\mathbb{E})$ the set of continuous functions f on $\mathbb{E}$ that tend to zero at infinity, we make the following assumption:

Assumption 1. *S is a Feller process on $\mathbb{E}$, that is, for any $f \in C_0(\mathbb{E})$, the family $(P_t f)_{t \geq 0}$, with $P_t f$ defined in (2.3), satisfies:*

- (i) $P_t f \in C_0(\mathbb{E})$ for any $t > 0$;
- (ii) $\lim_{t \rightarrow 0} P_t f(x) = f(x)$ for any $x \in \mathbb{E}$.

The Feller property guarantees that there exists a version of the process S with càdlàg paths satisfying the strong Markov property. In particular, a Feller process is a Hunt process.

Throughout the paper we will take the knock-out set A to be of the form

$$(2.5) \quad A = [0, \ell] \cup [u, \infty), \quad 0 \leq \ell < u \leq \infty,$$

which includes the cases of double and single barrier options—the latter by taking $\ell = 0$ or $u = \infty$. To rule out degeneracies we will make the following assumption on the behaviour of S at the boundary points ℓ and u :

Assumption 2. For every $x \in \mathbb{E}$ we have $\mathbf{P}_x(\tau_A = \tau_{A^o}) = 1$ where $A^o = [0, \ell) \cup (u, \infty)$.

This assumption states that the first entrance times into A and its interior coincide almost surely, if the spot S_0 is equal to x . If $u < \infty$, a sufficient condition for Assumption 2 to be satisfied is $\mathbf{P}_x(\tau_{A^o} = 0) = 1$ for $x \in \{\ell, u\}$; that is, when started at ℓ or u , the process S immediately enters A^o . The class of Feller processes satisfying Assumption 2 includes many of the models employed in quantitative finance such as (Feller-)diffusions, jump-diffusions with non-generate diffusion coefficient and Lévy processes whose Lévy measure admits a density.

The family $(P_t)_{t \geq 0}$ is determined by its infinitesimal generator $\mathcal{L}$ that is defined as

$$(2.6) \quad \mathcal{L}f(x) := \lim_{t \downarrow 0} \frac{1}{t}(P_t f - f)(x)$$

for any function $f \in C_0(\mathbb{E})$ for which the right-hand side of (2.6) converges in the strong sense.¹ The set $\mathcal{D}$ of such functions is called the domain of the operator $\mathcal{L}$ and is dense in $C_0(\mathbb{E})$. These fundamental facts about semigroups and their generators can be found in [23, Ch. 1].

We next give a few examples of Feller processes with their generators.

Example 1. A diffusion asset price model $S = \{S_t\}_{t \geq 0}$ evolves under a risk-neutral measure according to the stochastic differential equation (SDE)

$$(2.7) \quad \frac{dS_t}{S_t} = \gamma dt + \sigma(S_t) dW_t,$$

where $S_0 > 0$ is the initial price, $\gamma \in \mathbb{R}$ and $\sigma : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a given measurable function. To guarantee the absence of arbitrage we assume that σ is chosen such that the discounted process $\{e^{-\gamma t} S_t\}_{t \geq 0}$ is a martingale, which implies in particular that S does not explode to infinity. If, in addition, infinity is not entrance² for S and σ is a continuous function, then S is a Feller process, and its infinitesimal generator $\mathcal{L}_D$ acts on $f \in C_c^2(\mathbb{E})$ ³ as

$$(2.8) \quad \mathcal{L}_D f(x) = \frac{\sigma(x)^2 x^2}{2} f''(x) + \gamma x f'(x),$$

where f' denotes the derivative of f with respect to x (see [23, Sec. 8.1]).

Example 2. (a) The price process in an exponential Lévy model S given by

$$(2.9) \quad S_t := S_0 e^{(r-d)t} \frac{e^{L_t}}{\mathbf{E}[e^{L_t}]}$$

where r and d are constants representing the interest rate and dividend yield and $L = \{L_t\}_{t \geq 0}$ is a Lévy process, such that $\mathbf{E}[e^{L_t}] < \infty$ for all $t > 0$. By construction, $\{e^{-(r-d)t} S_t\}_{t \geq 0}$ is a martingale. Further, $\mathbf{E}[e^{L_t}] < \infty$ if and only if the Lévy measure ν integrates $\exp(y)$ at infinity, that is,

$$(2.10) \quad \int_{(1, \infty)} e^y \nu(dy) < \infty.$$

¹That is, the convergence is with respect to the norm $\|f\| := \sup_{x \in \mathbb{E}} |f(x)|$ of the Banach space $(C_0(\mathbb{E}), \|\cdot\|)$.

²See Itô and McKean [35] for an explicit criterion in terms of γ and σ for this to be the case.

³ $C_c^2(\mathbb{E})$ denotes the set of C^2 functions with compact support in $\mathbb{E}^o = (0, \infty)$.

The law of L is determined by its characteristic exponent Ψ , which is related to the characteristic function Φ_t of L_t by $\Phi_t(s) = \exp(t\Psi(s))$ and which, under condition (2.10), has the Lévy-Khintchine representation

$$\Psi(s) = \mathbf{i}cs - \frac{\sigma^2 s^2}{2} + \int_{\mathbb{R}} (e^{\mathbf{i}sy} - 1 - \mathbf{i}sy) \nu(dy),$$

where (c, σ^2, ν) is the characteristic triplet, with $\sigma, c \in \mathbb{R}$ and ν the Lévy measure, which satisfies the integrability condition $\int_{(-1,1)} |y|^2 \nu(dy) < \infty$. The process S is a Feller process with an infinitesimal generator acting on $f \in C_c^2(\mathbb{E})$ as (cf. Sato [57, Thm. 31.5])

$$\mathcal{L}f(x) = \frac{\sigma^2 x^2}{2} f''(x) + \gamma x f'(x) + \int_{\mathbb{R}} [f(xe^y) - f(x) - xf'(x)(e^y - 1)] \nu(dy),$$

where

$$\gamma = r - d.$$

(b) A closely related model is a geometric Lévy process specified by the SDE

$$\frac{dS_t}{S_{t-}} = (r - d - c)dt + dL_t, \quad t > 0,$$

where $S_0, \mu \in (0, \infty)$ and L is a Lévy process with characteristic triplet (c, σ^2, ν) , where the Lévy measure ν has support in $(-1, \infty)$ and

$$\int_{(1, \infty)} z \nu(dz) < \infty.$$

The former condition guarantees that $S_t > 0$ for all $t > 0$, the latter that $\mathbf{E}[|L_t|] < \infty$ for all $t > 0$. The discounted process $\{e^{-(r-d)t} S_t\}_{t \geq 0}$ is a martingale, and S is a Feller process with an infinitesimal generator $\mathcal{L}$ that acts on $f \in C_c^2(\mathbb{E})$ as

$$\mathcal{L}f(x) = \frac{\sigma^2 x^2}{2} f'(x) + \gamma x f'(x) + \int_{(-1, \infty)} [f(x(1+z)) - f(x) - xf'(x)z] \nu(dz),$$

where $\gamma = r - d$. For further background on Lévy processes we refer to Sato [57].

Example 3. More generally, one may specify the price process S by directly prescribing its generator $\mathcal{L}$ to act on sufficiently regular functions f as

$$(2.11) \quad \mathcal{L}f(x) = \mathcal{L}_D f(x) + \mathcal{L}_J f(x),$$

where $\mathcal{L}_D f$ is given in (2.8) and

$$\mathcal{L}_J f(x) = \int_{(-1, \infty)} [f(x(1+y)) - f(x) - f'(x)xy] \nu(x, dy),$$

where for every $x \in \mathbb{E}$, $\nu(x, dy)$ is a (Lévy) measure with support in $(-1, \infty)$ such that

$$\int_{(-1, \infty)} \min\{y^2, |y|\} \nu(x, dy) < \infty.$$

The discounted process $\{e^{-\gamma t} S_t\}_{t \geq 0}$ is a local martingale. Sufficient conditions on σ and ν to guarantee the existence of a Feller process S corresponding to this generator were established in Kolokoltsov [38, Thm. 1.1].

A key step in solving the barrier valuation problems is the observation that the expected values of knock-out options and general barrier options can be expressed in terms of the *marginal* distributions of two Markov processes associated to S . Given a Markov process S the processes that have the same dynamics as S before entering A , but are stopped or jump to the graveyard state ∂ upon entering the set A , respectively, are themselves Markov processes. Let $\hat{S}^A = \{S_t \mathbf{I}_{\{t < \tau_A\}} + \partial \mathbf{I}_{\{t \geq \tau_A\}}\}_{t \geq 0}$ denote the killed process and $\tilde{S}^A$ the process killed at rate r that is stopped upon entering the set A . Then the value of the barrier options can be expressed as

$$\begin{aligned} \mathbf{E}_x [g(S_T) \mathbf{I}_{\{\tau_A > T\}}] &= \mathbf{E}_x [g(\hat{S}_T^A)] =: \hat{P}_T^A g(x), \\ \mathbf{E}_x [e^{-rT} g(S_T) \mathbf{I}_{\{\tau_A > T\}}] + \mathbf{E}_x [e^{-r\tau_A} h(S_{\tau_A}) \mathbf{I}_{\{\tau_A \leq T\}}] &= \mathbf{E}_x [e^{-r(T \wedge \tau_A)} f(S_{T \wedge \tau_A})] \\ &= \mathbf{E}_x [f(\tilde{S}_T^A)] =: \tilde{P}_T^A f(x), \end{aligned}$$

where we assume that $g(\partial) = 0$ and the function f is defined as

$$f(x) = \begin{cases} h(x), & x \in A, \\ g(x), & x \notin A. \end{cases}$$

To calculate the value-functions of barrier options written on the underlying price process S we thus need to identify $\hat{P}_T^A g(x)$ and $\tilde{P}_T^A f(x)$. This can be achieved by employing the infinitesimal generators of the semigroups $(\hat{P}_t^A)_{t \geq 0}$ and $(\tilde{P}_t^A)_{t \geq 0}$ associated to the Markov processes $\hat{S}^A$ and $\tilde{S}^A$ which are explicitly expressed in terms of the generator $\mathcal{L}$ as follows:

Lemma 1. (i) For any $f \in \mathcal{D}$, where $\mathcal{D}$ is the domain of the generator $\mathcal{L}$, we have

$$(2.12) \quad \lim_{t \downarrow 0} t^{-1} (\tilde{P}_t^A f(x) - f(x)) = \tilde{k}(x) := \begin{cases} 0, & x \in A, \\ (\mathcal{L} - r)f(x), & x \notin A, \end{cases}$$

where the convergence is pointwise. If $\tilde{S}^A$ is a Feller process, and

$$(2.13) \quad \lim_{x \rightarrow \partial A} \{\mathcal{L}f(x) - rf(x)\} = 0, \quad \text{where } \partial A \text{ is the boundary of } A,$$

then $\tilde{\mathcal{L}}^A f = \tilde{k}$, where $\tilde{\mathcal{L}}^A$ is the infinitesimal generator of the semigroup $(\tilde{P}_t^A)_{t \geq 0}$.

(ii) Let $g \in \mathcal{D}$ and assume that x satisfies

$$(2.14) \quad \mathbf{P}_x(\tau_A \leq t) = o(t) \text{ as } t \searrow 0 \quad \text{or} \quad g(S_{\tau_A}) = 0 \quad \mathbf{P}_x\text{-a.s.}$$

Then

$$(2.15) \quad \lim_{t \downarrow 0} t^{-1} (\hat{P}_t^A g(x) - g(x)) = \mathcal{L}g(x),$$

where the convergence is pointwise. If $\hat{S}^A$ is a Feller process and

$$g|_A = 0 \quad \text{and} \quad \lim_{x \rightarrow \partial A} \mathcal{L}g(x) = 0,$$

then $\hat{\mathcal{L}}^A g = \mathcal{L}g$, where $\hat{\mathcal{L}}^A$ is the infinitesimal generator of the semigroup $(\hat{P}_t^A)_{t \geq 0}$.

The proof of Lemma 1 is given in the Appendix. If $\tilde{S}^A$ and $\hat{S}^A$ are themselves Feller processes, the relations between $\tilde{P}^A$ and $\tilde{\mathcal{L}}^A$, and between $\hat{P}^A$ and $\hat{\mathcal{L}}^A$ can formally be expressed as follows:

$$(2.16) \quad \tilde{P}_t^A = \exp\left(t\tilde{\mathcal{L}}^A\right), \quad \hat{P}_t^A = \exp\left(t\hat{\mathcal{L}}^A\right).$$

Equation (2.16) can be given a precise meaning if, for example, $\tilde{P}_t^A$ and $\hat{P}_t^A$ can be defined as a self-adjoint operator on a separable Hilbert space (see e.g. Ch. XII in Dunford and Schwarz [20], or Hille and Philips [33]). By determining the spectral decompositions of $\tilde{\mathcal{L}}^A$ and $\hat{\mathcal{L}}^A$ one can construct spectral expansions of $\tilde{P}_t^A f(x)$ and $\hat{P}_t^A f(x)$, which in the case of a discrete spectrum reduces to a series expansion. See Linetsky [45, 46, 47] for a development of this spectral expansion approach for one-dimensional diffusion models in finance, and an overview of related literature.

When (asymmetric) jumps are present, the operator is non-local and not self-adjoint, and the spectral theory has been less well developed, with fewer explicit results. Here we will follow a different approach: we will approximate S by a finite state Markov chain, and show that for the approximating chain a matrix analog of the identities (2.12)–(2.16) holds true, where the infinitesimal generators $\tilde{\mathcal{L}}^A$ and $\hat{\mathcal{L}}^A$ can be easily obtained from $\mathcal{L}$. We give a self-contained development of this approach in Section 3, and present an extension to time-dependent dynamics in Section 4.

3. EXIT PROBABILITIES FOR CONTINUOUS-TIME MARKOV CHAINS

Given a Markov price process S of interest, the idea is to construct a continuous-time finite state Markov chain X whose dynamics are “close” to those of S , and to calculate the relevant expectations for this approximating chain. In this section we will focus on the latter; we will return to the question of how to construct such a chain in Section 5. Assume therefore we are given a finite state continuous-time Markov chain $X = \{X_t\}_{t \geq 0}$. From Markov chain theory it is well known that the chain is completely specified by its state-space (or grid) $\mathbb{G} \subset \mathbb{E}$ and its generator matrix Λ , which is an $N \times N$ square matrix with zero row sums and non-positive diagonal elements, if $\mathbb{G}$ has N elements. Given the generator matrix Λ the family of transition matrices $(P_t)_{t \geq 0}$ of X , defined by $P_t(x, y) := \mathbf{P}_x[X_t = y]$ for $x, y \in \mathbb{G}$, is given by

$$P_t = \exp(t\Lambda).$$

In particular, the expected discounted pay-off $\phi(X_T)$ at maturity T is then given by

$$(3.1) \quad \mathbf{E}_x[e^{-rT}\phi(X_T)] = e^{-rT} \cdot (\exp(T\Lambda)\phi)(x)$$

for $x \in \mathbb{G}$ and any function $\phi : \mathbb{G} \rightarrow \mathbb{R}$. Here and throughout the paper we will identify any square matrix $\mathcal{A} \in \mathbb{R}^{N \times N}$ and any vector ϕ in $\mathbb{R}^N$ with functions

$$\begin{aligned} \mathcal{A} : \mathbb{G} \times \mathbb{G} &\rightarrow \mathbb{R}, & \mathcal{A}(x, y) &:= e'_x \mathcal{A} e_y, & x, y &\in \mathbb{G}, \quad \text{and} \\ \phi : \mathbb{G} &\rightarrow \mathbb{R}, & \phi(x) &:= e'_x \phi, & x &\in \mathbb{G}, \end{aligned}$$

where the vectors e_x, e_y denote the corresponding standard basis vectors of $\mathbb{R}^N$ and $'$ stands for transposition.

The generator Λ can be retrieved from the family of matrices $(P_t)_{t \geq 0}$ defined above by differentiation at $t = 0$, that is,

$$P_t = I + t\Lambda + o(t) \quad \text{as } t \searrow 0.$$

In order for X itself to form a pricing model, defined under a martingale measure, we will suppose in addition that X is non-negative and that the discounted process is a martingale; more precisely we assume that $\{M_t\}_{t \geq 0}$ is a martingale where

$$(3.2) \quad M_t = \begin{cases} e^{-\gamma t} X_t, & t < \zeta, \\ e^{-\gamma \zeta} X_\zeta, & t \geq \zeta, \end{cases}$$

with $\zeta = \inf\{t \geq 0 : X_t \in \partial\mathbb{G}\}$ the hitting time of the “boundary” $\partial\mathbb{G}$ which consists of the smallest and largest elements of the state space $\mathbb{G}$, and $\gamma = r - d$. The Markov property of X implies that the process M given in (3.2) is a martingale precisely if

$$(3.3) \quad (\Lambda\eta)(x) = \gamma\eta(x) \quad \text{for } x \in \mathbb{G} \setminus \partial\mathbb{G},$$

where the function $\eta : \mathbb{G} \rightarrow \mathbb{E}$ is given by $\eta(z) = z$. Below we will show how to express the exit probabilities of the chain using matrix exponentiation in a way that is identical in form to the expected value (3.1) of a European pay-off. To that end, we partition $\mathbb{G}$ into a ‘continuation’ set $\widehat{\mathbb{G}}$ and a ‘knock-out’ set $\widehat{\mathbb{G}}^c := \mathbb{G} \setminus \widehat{\mathbb{G}}$, where

$$(3.4) \quad \widehat{\mathbb{G}} := \{x \in \mathbb{G} : x \in A^c\},$$

and define the first exit time of X from $\widehat{\mathbb{G}}$ by

$$(3.5) \quad \tau := \inf\{t \in \mathbb{R}_+ : X_t \notin \widehat{\mathbb{G}}\},$$

where we use the convention $\inf \emptyset := \infty$ and where we will take the set A as in (2.5).

The value of a general barrier knock-out option with a rebate depends on the joint distribution of the exit time τ_A from A and the positions of the underlying at maturity and at the moment of exit. The corresponding quantities for the chain X can be expressed in terms of two transformations of X , namely the chain $\widehat{X}$ that is killed upon exiting $\widehat{\mathbb{G}}$ and the chain $\widetilde{X}$ that is absorbed at that instance, respectively. Correspondingly, we associate to the generator matrix Λ , two matrices: the $\widehat{N} \times \widehat{N}$ matrix $\widehat{\Lambda}$, where $\widehat{N} := |\widehat{\mathbb{G}}|$, and the $N \times N$ matrix $\widetilde{\Lambda}_r$, defined by

$$(3.6) \quad \widetilde{\Lambda}_r(x, y) := \begin{cases} \Lambda(x, y) - r & \text{if } x \in \widehat{\mathbb{G}}, x = y, \\ \Lambda(x, y) & \text{if } x \in \widehat{\mathbb{G}}, y \in \mathbb{G}, x \neq y, \\ 0 & \text{if } x \in \widehat{\mathbb{G}}^c, y \in \mathbb{G}, \end{cases}$$

$$(3.7) \quad \widehat{\Lambda}(x, y) := \widehat{\Lambda}_0(x, y) := \Lambda(x, y) \quad \text{if } x \in \widehat{\mathbb{G}}, y \in \widehat{\mathbb{G}}.$$

We can now state the key result of this section:

Theorem 1. *For any $T > 0$, $x \in \mathbb{G}$ and $r \geq 0$ and any function $\phi : \mathbb{G} \rightarrow \mathbb{R}$ it holds that*

$$(3.8) \quad \mathbf{E}_x \left[e^{-r(T \wedge \tau)} \phi(X_{T \wedge \tau}) \right] = \left(\exp \left(T \widetilde{\Lambda}_r \right) \phi \right) (x).$$

In particular, for $\psi : \widehat{\mathbb{G}} \rightarrow \mathbb{R}$ and $\xi : \mathbb{G} \rightarrow \mathbb{R}$ with $\xi(x) = 0$ for $x \in \widehat{\mathbb{G}}$ we have that

$$(3.9) \quad \mathbf{E}_x[\psi(X_T)\mathbf{I}_{\{\tau > T\}}] = \left(\exp\left(T\widehat{\Lambda}\right)\psi\right)(x) \quad \text{for any } x \in \widehat{\mathbb{G}},$$

$$(3.10) \quad \mathbf{E}_x[\phi(X_T)\mathbf{I}_{\{\tau \leq T\}}] = (\exp(\Lambda T)\phi)(x) - \left(\exp\left(T\widehat{\Lambda}\right)\widehat{\phi}\right)(x)\mathbf{I}_{\widehat{\mathbb{G}}}(x) \quad \text{for any } x \in \mathbb{G},$$

$$(3.11) \quad \mathbf{E}_x[e^{-r\tau}\xi(X_\tau)\mathbf{I}_{\{\tau \leq T\}}] = \left(\exp\left(T\widetilde{\Lambda}_r\right)\xi\right)(x) \quad \text{for any } x \in \mathbb{G},$$

where $\widehat{\phi} = \phi|_{\widehat{\mathbb{G}}}$, the restriction of ϕ to $\widehat{\mathbb{G}}$.

Formulas (3.9)–(3.11) give a simple way of computing barrier option prices by a single matrix exponentiation. The expectation in (3.9) can be obtained by computing the spectral decomposition of the matrix $\widehat{\Lambda} = UDU^{-1}$ and applying the formula $\exp\left(T\widehat{\Lambda}\right) = U \exp(TD)U^{-1}$. The powerful Padé approximation method for matrix exponentiation, described in [32], can also be used to compute efficiently the matrix exponentials in Theorem 1. Since the state-space is finite, Theorem 1 is a corollary of Lemma 1. We present next a direct probabilistic derivation.

Proof. To prove equation (3.8), we will verify that the expected value of an Arrow-Debreu barrier security that pays 1 precisely if X is in the state y at the earlier of the maturity T and the knock-out time τ is given by

$$(3.12) \quad \mathbf{E}_x[e^{-r(T \wedge \tau)}\mathbf{I}_{\{X_{T \wedge \tau} = y\}}] = \left(\exp(T\widetilde{\Lambda}_r)\right)(x, y) \quad \text{for all } x, y \in \mathbb{G}.$$

For a given time grid $\mathbb{T}_n = (k\Delta t, k = 0, 1, 2, \dots, n)$ with $\Delta t = T/n$ denote by $\widetilde{P}_T^{(n)}(x, y)$ the expected value of the corresponding discretely monitored Arrow-Debreu security and let

$$\tau_n = \inf\{s \in \mathbb{T}_n : X_s \notin \widehat{\mathbb{G}}\}$$

be the corresponding time at which the barrier is crossed. Since the paths of the chain X are piecewise constant, it follows that $\tau_n \downarrow \tau$ and $X_{\tau_n} \rightarrow X_\tau$ as n tends to infinity. Hence the expected values of the discretely monitored Arrow-Debreu securities converge to the expected value of the continuously monitored one,

$$\widetilde{P}_T^{(n)}(x, y) = \mathbf{E}_x[e^{-r(T \wedge \tau_n)}\mathbf{I}_{\{X_{T \wedge \tau_n} = y\}}] \longrightarrow \mathbf{E}_x[e^{-r(T \wedge \tau)}\mathbf{I}_{\{X_{T \wedge \tau} = y\}}].$$

Clearly, since $\widehat{\mathbb{G}}^c$ is the knock-out set, it holds for all $t \geq 0$ that

$$\begin{aligned} \widetilde{P}_t^{(n)}(x, y) &= \begin{cases} 1 & \text{if } x \in \widehat{\mathbb{G}}^c, x = y \\ 0 & \text{if } x \in \widehat{\mathbb{G}}, x \neq y \end{cases} \\ &= (I - \bar{I})(x, y) \quad \text{for all } x \in \widehat{\mathbb{G}}^c, y \in \mathbb{G}, \end{aligned}$$

where $\bar{I}$ is a square matrix of size N with $\bar{I}(x, x) = 1$ if $x \in \widehat{\mathbb{G}}$ and zero else. Further, for $x \in \widehat{\mathbb{G}}$ an application of the Markov property of X shows that

$$\begin{aligned} \widetilde{P}_T^{(n)}(x, y) &= e^{-r\Delta t} \sum_{z \in \mathbb{G}} P_{\Delta t}(x, z) \widetilde{P}_{T-\Delta t}^{(n)}(z, y) \\ &= \left(\bar{I} (e^{-r\Delta t} P_{\Delta t}) \widetilde{P}_{T-\Delta t}^{(n)}\right)(x, y). \end{aligned}$$

Combining the two cases, iterating the argument and using the differentiability of P_t at $t = 0$ shows that

$$\begin{aligned}\tilde{P}_T^{(n)}(x, y) &= \left((I - \bar{I} + \bar{I} e^{-r\Delta t} P_{\Delta t}) \tilde{P}_{T-\Delta t}^{(n)} \right)(x, y) \\ &= \left((I - \bar{I} + \bar{I} e^{-r\Delta t} P_{\Delta t})^{T/\Delta t} \right)(x, y) \\ &= \left((I + \bar{I} e^{-r\Delta t} (P_{\Delta t} - I) + \bar{I} (e^{-r\Delta t} - 1))^{T/\Delta t} \right)(x, y) \\ &= \left((I + \Delta t (\tilde{\Lambda}_0 - r\bar{I}) + o(\Delta t))^{T/\Delta t} \right)(x, y),\end{aligned}$$

since $\tilde{\Lambda}_0 = \bar{I}\Lambda$. When $\Delta t = T/n$ tends to zero, this expression converges to $\left(\exp(T\tilde{\Lambda}_r) \right)(x, y)$, which completes the proof of (3.12) and hence implies (3.8). Equation (3.11) follows then directly by applying (3.8) to ξ . Noting that for any $\psi : \hat{\mathbb{G}} \rightarrow \mathbb{R}$ and any $x \in \hat{\mathbb{G}}$ we have

$$\left(\hat{\Lambda}\psi \right)(x) = \left(\tilde{\Lambda}_0\psi_0 \right)(x), \quad \text{where } \psi_0 : \mathbb{G} \rightarrow \mathbb{R} \text{ is given by } \psi_0(y) := \psi(y)\mathbf{I}_{\hat{\mathbb{G}}}(y),$$

we get that

$$\left(\exp(T\hat{\Lambda})\psi \right)(x) = \left(\exp(T\tilde{\Lambda}_0)\psi_0 \right)(x),$$

which yields (3.9). Finally, (3.10) is a direct consequence of (3.1), (3.9) and the fact that a European option with pay-off $\phi(X_T)$ is equal to the sum of a knock-out barrier option and a knock-in barrier option with the same pay-off $\phi(X_T)$ and same knock-out/knock-in levels. $\square$

4. TIME-INHOMOGENEOUS MARKOV PROCESSES

In practical applications model parameters such as the short rate or the volatility function are often taken to be time-dependent. In this section we will show how the approach developed in Sections 2 and 3 can be adapted to this time-dependent setting.

Let a time-inhomogeneous Markov process $S = \{S_t\}_{t \in [0, T]}$, model the evolution of the risky asset price until some maturity $T > 0$. To highlight the role of time as a factor, we consider the time-space Markov process (D, Y) where $D = \{D_t\}_{t \geq 0}$ and $Y = \{Y_t\}_{t \geq 0}$ are defined by

$$(4.1) \quad D_t := (D_0 + t) \wedge T \quad \text{and} \quad Y_t := S_{D_t} \quad t \geq 0, D_0 \in [0, T].$$

Then, for any $s, t > 0$ and bounded Borel function $f : [0, T] \times \mathbb{E} \rightarrow \mathbb{R}$ it holds that

$$\mathbf{E}[f(D_t, Y_t) | \mathcal{F}_s] = (Q_t f)(D_s, Y_s),$$

where

$$\begin{aligned}(Q_t f)(s, x) &= \mathbf{E}_{s, x}[f(D_t, Y_t)] := \mathbf{E}[f(D_t, Y_t) | D_0 = s, Y_0 = x] \\ &= \mathbf{E}[f((s + t) \wedge T, S_{(s+t) \wedge T}) | S_s = x].\end{aligned}$$

Thus the family $(Q_t)_{t \geq 0}$ forms a (two-dimensional) semigroup. We will restrict ourselves to the case when (D, Y) is a Feller process with state-space $\mathbb{E}' := [0, T] \times \mathbb{E}$, in the sense of Assumption 1

with $\mathbb{E}$ replaced by $\mathbb{E}'$. Associated to the semigroup $(Q_t)_{t \geq 0}$ is the infinitesimal generator $\mathcal{L}'$ defined by

$$\lim_{t \downarrow 0} t^{-1}[Q_t f(s, x) - f(s, x)] = \mathcal{L}' f(s, x),$$

for all f in the set $\mathcal{D}'$, the domain of $\mathcal{L}'$, for which this limit exists. The process stopped upon entering a closed set A remains a Markov process, and its corresponding generator can be obtained as in Section 2. Denote by $\mathcal{L}_t$ the generator restricted to functions $g : \mathbb{E} \rightarrow \mathbb{R}$ of space only, that is, $(\mathcal{L}_t g)(x) := (\mathcal{L}' g)(t, x)$.

We next give two examples of models in which time-dependence plays a natural role — we will present a numerical illustration of barrier valuation under these models in Section 7.

Example 4. Given a surface of arbitrage-free call option prices generated by some diffusion model, it is well known that there exists a one-dimensional diffusion with time-dependent volatility function $\Sigma(t, x)$ that reproduces those option prices. The volatility function $\Sigma(t, x)$ is explicitly given in terms of the call option prices by Dupire's formula. In such a local volatility model the stock price $S = \{S_t\}_{t \in [0, T]}$ evolves according to the SDE

$$(4.2) \quad dS_t = \gamma(t, S_t)dt + \Sigma(t, S_t)dW_t,$$

where $\gamma(t, x) = (r(t) - d(t))x$ with $r(t)$ and $d(t)$ continuous functions representing the interest and dividend rate, and $\Sigma(t, x)$ is a volatility function. If the latter can be chosen sufficiently regular, $\{(D_t, Y_t)\}_{t \geq 0}$ will be a Feller process and the discounted stock price $\{e^{-\int_0^t [r(s) - d(s)]ds} S_t\}_{t \in [0, T]}$ a martingale. The corresponding infinitesimal generator then acts on functions $f \in C_c^2(\mathbb{E})$ as

$$(\mathcal{L}_t f)(x) = \gamma(t, x)f'(x) + \frac{1}{2}\Sigma^2(t, x)f''(x).$$

Example 5. In an exponential additive model the log-price is a spatially homogeneous Markov process, and $S = \{S_t\}_{t \in [0, T]}$ is given by

$$(4.3) \quad S_t = e^{\int_0^t [r(s) - d(s)]ds} S_0 \frac{e^{X_t}}{\mathbf{E}[e^{X_t}]},$$

where X is an additive process, a process with independent increments and càdlàg paths that starts at zero. If $\mathbf{E}[e^{X_t}] < \infty$ for all $t \in [0, T]$, the discounted process $\{e^{-\int_0^t [r(s) - d(s)]ds} S_t\}_{t \in [0, T]}$ is a martingale, by construction.

To be specific, let X be an additive process $X = \{X_t\}_{t \in [0, T]}$ with drift and volatility functions $\beta(t)$ and $\sigma(t)$, and a system of Lévy measures $\nu(t, dy)$, given by

$$(4.4) \quad X_t = \int_0^t \beta(s)ds + \int_0^t \sigma(s)dW_s + Z_t^\nu.$$

Here the process $Z^\nu = \{Z_t^\nu\}_{t \in [0, T]}$ is a jump-process with compensator $\nu(t, dy)dt = g(t, y)dydt$ satisfying the integrability condition

$$(4.5) \quad \int_{\mathbb{R} \setminus \{0\}} \min\{|e^y - 1|, y^2\} \sup\{g(s, y) : s \in [0, T]\} dy < \infty,$$

with $g(\cdot, y) : [0, T] \rightarrow \mathbb{R}_+$, $y \in \mathbb{R} \setminus \{0\}$, a family of continuous functions. Models from this class (with $\sigma \equiv 0$ and a particular form of $g(t, y)$) have been proposed for stock price modelling by Carr

et al. [12] — see also Section 7.3.2. Then the time-space process $\{(D_t, Y_t)\}_{t \geq 0}$ is a Feller process, with an infinitesimal generator acting on $f \in C_c^2(\mathbb{E})$ as⁴

$$(4.6) \quad \begin{aligned} (\mathcal{L}_t f)(x) &= \gamma(t) x f'(x) + \frac{\sigma^2(t)}{2} x^2 f''(x) \\ &+ \int_{\mathbb{R} \setminus \{0\}} [f(xe^y) - f(x) - f'(x)x(e^y - 1)] g(t, y) dy, \end{aligned}$$

where $\gamma(t) = r(t) - d(t)$.

4.1. Time-inhomogeneous chains. To approximate the barrier option prices driven by a given time-inhomogeneous Markov process we again start by constructing an appropriate Markov chain. We build an approximating time-inhomogeneous chain with a generator that is piecewise constant in time. Given a partition $\mathbb{T} = \{T_i^{(m)}\}_{i=0}^m$ with $T_0 = 0 < T_1^{(m)} < \dots < T_m^{(m)} = T$ of $[0, T]$, we assume that the chain X on the state-space $\mathbb{G}$ evolves according to the generator $\Lambda^{(i)}$ during the time-interval $[T_{i-1}^{(m)}, T_i^{(m)})$ where the matrices $\Lambda^{(i)}$ are chosen so as to approximate well the infinitesimal generators $\mathcal{L}_{T_i^{(m)}}$. Thus X has a time-dependent generator given by

$$(4.7) \quad \Lambda_t^{(m)} := \sum_{j=1}^m \Lambda^{(j)} \mathbf{I}_{[T_{j-1}^{(m)}, T_j^{(m)})}(t).$$

Also we take the short rate $r^{(m)}$ to be piecewise constant

$$(4.8) \quad r^{(m)}(t) = \sum_{i=1}^m r_i \mathbf{I}_{[T_{i-1}^{(m)}, T_i^{(m)})}(t), \quad \text{for } t \geq 0 \text{ and } r_i \geq 0, i = 1, \dots, m.$$

As a straightforward consequence of Theorem 1 and the Markov property we have the following result for the first-passage probabilities of the chain X .

Corollary 1. *For $r^{(m)}$ given as above and any function $\phi : \mathbb{G} \rightarrow \mathbb{R}$ it holds that*

$$(4.9) \quad \mathbf{E}_{0,x} \left[e^{-\int_0^{T \wedge \tau} r^{(m)}(s) ds} \phi(X_{T \wedge \tau}) \right] = \left(\exp \left(\Delta T_1 \tilde{\Lambda}_{r_1}^{(1)} \right) \cdots \exp \left(\Delta T_m \tilde{\Lambda}_{r_m}^{(m)} \right) \phi \right) (x).$$

In particular, for any functions $\psi : \hat{\mathbb{G}} \rightarrow \mathbb{R}$ and $\xi : \mathbb{G} \rightarrow \mathbb{R}$ with $\xi(x) = 0$ for $x \in \hat{\mathbb{G}}$, the following equalities hold

$$(4.10) \quad \mathbf{E}_{0,x} [\psi(X_T) \mathbf{I}_{\{\tau > T\}}] = \left(\exp \left(\Delta T_1 \hat{\Lambda}^{(1)} \right) \cdots \exp \left(\Delta T_m \hat{\Lambda}^{(m)} \right) \psi \right) (x), \quad x \in \hat{\mathbb{G}},$$

$$(4.11) \quad \begin{aligned} \mathbf{E}_{0,x} [\phi(X_T) \mathbf{I}_{\{\tau \leq T\}}] &= \left(\exp \left(\Delta T_1 \Lambda^{(1)} \right) \cdots \exp \left(\Delta T_m \Lambda^{(m)} \right) \phi \right) (x) \\ &- \left(\exp \left(\Delta T_1 \hat{\Lambda}^{(1)} \right) \cdots \exp \left(\Delta T_m \hat{\Lambda}^{(m)} \right) \hat{\phi} \right) (x) \mathbf{I}_{\hat{\mathbb{G}}}(x) \end{aligned}$$

$$(4.12) \quad \mathbf{E}_{0,x} \left[e^{-\int_0^\tau r^{(m)}(t) dt} \xi(X_\tau) \mathbf{I}_{\{\tau \leq T\}} \right] = \left(\exp \left(\Delta T_1 \tilde{\Lambda}_{r_1}^{(1)} \right) \cdots \exp \left(\Delta T_m \tilde{\Lambda}_{r_m}^{(m)} \right) \xi \right) (x), \quad x \in \mathbb{G},$$

where $\Delta T_i := T_i^{(m)} - T_{i-1}^{(m)}$, $\hat{\phi} = \phi|_{\hat{\mathbb{G}}}$ and $\hat{\Lambda}^{(i)}$ and $\tilde{\Lambda}_{r_i}^{(i)}$ are defined as in (3.7) and (3.6).

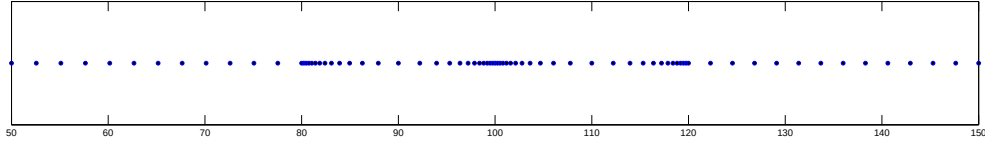


FIGURE 2. The figure shows a non-uniform grid that was generated using the algorithm described in Section 5. The grid consists of $N = 75$ points that are densely distributed around the spot $S_0 = 100$ and the upper and lower barriers $\ell = 80$ and $u = 120$; its smallest and largest values were chosen as $x_1 = 50$ and $x_N = 150$, and $N_i = 25$, $i = 1, 2, 3$, points were placed in the areas $[x_1, \ell]$, $[\ell, u]$ and $[u, x_N]$ with respective density parameters $d_1^- = 100$, $d_1^+ = 1$, $d_2^- = 1$, $d_2^+ = 1$, and $d_3^- = 1$, $d_3^+ = 100$.

5. CONSTRUCTION OF THE MARKOV CHAIN

The Markov chain approximation algorithm for homogeneous Feller processes can now be described as follows:

- (1) Construct an approximating Markov chain:
 - (a) specify a (non-uniform) grid $\mathbb{G}$;
 - (b) define a generator matrix Λ of a Markov chain with state-space $\mathbb{G}$.
- (2) Compute the barrier option prices using formulae (3.8)–(3.11).

In the case of a time-inhomogeneous Markov process this algorithm is modified as follows: first specify a partition $\mathbb{T} = \{T_i^{(m)}\}_{i=0}^m$ of the interval $[0, T]$ and apply step (1) of the algorithm above for every subinterval $[T_{i-1}^{(m)}, T_i^{(m)}]$, and then compute the barrier option prices using formulae (4.9) and (4.12).

A suitable choice of the grid in step (1a) is essential for the effectiveness of the above pricing algorithm. The construction of an optimal grid (according to some criterion) is a topic of separate study, which will not be pursued further in this paper. One of the features of a good grid is that it has sufficient resolution in regions of interest, such as the current spot value and the barrier levels, which is a necessary condition for constructing a Markov chain market model that approximates well the dynamics of the given price process. Another desirable feature is that the grid “covers” a sufficiently large part of the state space, which is needed to control the truncation error that arises when approximating an infinite state space by a finite state space. To employ a uniform grid that satisfies these conditions would be computationally expensive. An example of a procedure for generating a suitable non-uniform grid $\mathbb{G}$, based on an algorithm from [59], is as follows:

- (1) Pick $N_i \in \mathbb{N}$ and the density parameters $d_i^\pm \in (0, \infty)$, $i = 1, 2, 3$, and the smallest and largest values x_1, x_N of the grid $\mathbb{G}$, where $N = N_1 + N_2 + N_3$.
- (2) Define $\mathbb{G}_i := \text{GENERATESUBGRID}(a_i, s_i, b_i, N_i, d_i^-, d_i^+)$, for $i = 1, 2, 3$, where $a_1 = x_1, s_1 = \ell, b_1 = (S_0 + \ell)/2 = a_2, s_2 = S_0, b_2 = (u + S_0)/2 = a_3, s_3 = u, b_3 = x_N$.
- (3) Set $\mathbb{G} := \mathbb{G}_1 \cup \mathbb{G}_2 \cup \mathbb{G}_3$,

where the subgrid is generated by the following procedure:

⁴A proof of these facts is given in Appendix A.2.

GENERATESUBGRID(a, s, b, M, g_1, g_2)

- (1) Compute $c_1 = \operatorname{arcsinh}\left(\frac{a-s}{g_1}\right)$, $c_2 = \operatorname{arcsinh}\left(\frac{b-s}{g_2}\right)$.
- (2) Define the lower part of the grid by the formula $x_k := s + g_1 \sinh(c_1(1 - (k-1)/(M/2-1)))$ for $k \in \{1, \dots, M/2\}$. Note that $x_1 = a, x_{M/2} = s$.
- (3) Define the upper part of the grid using the formula $x_{k+M/2} := s + g_2 \sinh(c_2 2k/M)$ for $k \in \{1, \dots, M/2\}$. Note that $x_M = b$.

RETURN $\{x_1, \dots, x_M\}$

The non-uniform state-space $\mathbb{G} \subset \mathbb{E}$ for the Markov chain X is constructed by concatenating the three subgrids that are generated by specifying lower, middle and upper points $a < s < b$ and density parameters g_1 and g_2 . A smaller density corresponds to a grid that is more concentrated around the middle point s . Observe that the algorithm above places the current spot S_0 and the barrier levels $\ell < u$ on the grid $\mathbb{G}$ and that the resolution of the grid around ℓ, S_0 and u can be controlled by the density parameters $d_i^\pm, i = 1, 2, 3$. A Matlab implementation of this grid generator is given Appendix B. To download the code see [51].

The remainder of this section will be devoted to step (1b) in the algorithm described above. In Section 5.1 we review the well known case of diffusion processes; we consider Markov processes with discontinuous paths in Sections 5.2 and 5.3.

5.1. Diffusions. Let $S = (S_t)_{t \geq 0}$ be an asset price process which evolves under a risk-neutral measure according to the SDE in (2.7) of Example 1. For a given finite state-space $\mathbb{E}$ of the diffusion S (and a given barrier contract) we construct a non-uniform state-space $\mathbb{G}$ with N elements using the algorithm from the beginning of this section. Define the sets

$$\partial\mathbb{G} := \{x_1, x_N\} \quad \text{and} \quad \mathbb{G}^o := \mathbb{G} \setminus \partial\mathbb{G},$$

where the “boundary” $\partial\mathbb{G}$ consist of the smallest (i.e. x_1) and largest (i.e. x_N) elements in $\mathbb{G}$ and the “interior” $\mathbb{G}^o$ is the complement of the boundary.

The next task is to construct the approximating continuous-time Markov chain $X = (X_t)_{t \geq 0}$ on the state-space $\mathbb{G}$ by specifying its generator matrix Λ in such a way that the first and the second instantaneous moments of the processes S and X coincide on the set $\mathbb{G}^o$. In other words the following conditions need to be satisfied:

$$(5.1) \quad \mathbf{E}_x [(S_{\Delta t} - S_0)^j] = \mathbf{E}_x [(X_{\Delta t} - X_0)^j] + o(\Delta t), \quad \text{for } x \in \mathbb{G}^o, j \in \{1, 2\}.$$

This leads to the following system of equations for the elements of the matrix Λ . For each $x \in \mathbb{G}^o$ we have:

$$(5.2) \quad \sum_{y \in \mathbb{G}} \Lambda(x, y) = 0 \quad \text{and} \quad \Lambda(x, y) \geq 0 \quad \forall y \in \mathbb{G} \setminus \{x\},$$

$$(5.3) \quad \sum_{y \in \mathbb{G}} \Lambda(x, y)(y - x) = \gamma x,$$

$$(5.4) \quad \sum_{y \in \mathbb{G}} \Lambda(x, y)(y - x)^2 = \sigma(x)^2 x^2.$$

For $x \in \partial\mathbb{G}$ we impose an absorbing boundary condition: $\Lambda(x, y) = 0$ for all $y \in \mathbb{G}$. The condition in (5.2) ensures that Λ is a generator matrix and equation (5.3) implies that the chain X drifts locally at the same rate as the diffusion S .

In applications it is typically possible to find a tri-diagonal generator matrix Λ that satisfies the linear system in (5.2) – (5.4). In terms of the process X this implies that at any moment in time the chain can only jump to neighbouring states. By setting the top and bottom rows of the matrix Λ to zero, we make the states in the set $\partial\mathbb{G}$ absorbing. This behaviour of the chain X clearly differs from the dynamics of the diffusion S in the neighbourhood of the set $\partial\mathbb{G}$. To ensure that the laws of the processes X and S are close to each other during the finite time interval of interest we therefore have to choose the boundary states sufficiently far. Such a choice is computationally inexpensive because the state-space $\mathbb{G}$ is non-uniform. In practical applications we can ensure easily that the accumulation of probability mass in the states of $\partial\mathbb{G}$ is negligible during the time interval of interest.

In Sections 7.1.1 and 7.1.2 we compare numerical results of our algorithm with the corresponding results for the geometric Brownian motion (reported in [26] and [41]) and the CEV process (reported in [17]) respectively.

5.2. Lévy subordinated diffusions. A common way of building models with jumps is by time-changing diffusions with a Lévy subordinator (see e.g. 7.2.1). If the jump process has this special structure, its generator is closely related to the generator of the diffusion. In this section we recall how this relationship can be exploited to obtain a continuous-time Markov chain approximation of the jump process that admits this special structure.

Let S' be a time-homogenous diffusion governed by SDE (2.7) that satisfies the conditions in Example 1 and let Z be a Lévy subordinator (i.e. a Lévy process with non-decreasing paths that starts at 0) independent of S' . In this section we assume that the asset price process is modelled by the process $S = \{S_t\}_{t \geq 0}$, where

$$(5.5) \quad S_t := e^{\mu t} S'_{Z_t} \quad \text{for some } \mu \in \mathbb{R}.$$

It follows from the *Phillips theorem* (see [57, Thm. 32.1] or [52]) that S is a Markov process that satisfies the Feller property. Conditioning on the independent subordinator Z yields that the discounted asset price process $S = \{e^{-(r-d)t} S_t\}_{t \geq 0}$ is a martingale if and only if $\mathbf{E}_s[S_t] = e^{(r-d)t} s$ for all $t > 0$ and $s \in \mathbb{E}$, which is equivalent to

$$(5.6) \quad \mu + \psi_Z(-\gamma) = r - d$$

where ψ_Z is the Laplace exponent of the subordinator Z (i.e. $\mathbf{E}[e^{-uZ_t}] = e^{t\psi_Z(u)}$), γ is the drift in SDE (2.7) and r, d are the interest rate and the dividend yield respectively. The identity in (5.6) follows by conditioning on the independent subordinator Z and using the fact that $\mathbf{E}_s[S'_t] = se^{\gamma t}$ for all $t > 0$ and $s \in \mathbb{E}$. If (5.6) is satisfied, we can use S as a model for the evolution of the arbitrage free price of risky asset price under a risk-neutral measure.

We will now use the structure of the process S given by subordination (5.5) to define a generator matrix for the Markov chain that approximates the asset price process S . Recall first that the

generator $\mathcal{G}'$ of the diffusion S' can be expressed as

$$(\mathcal{G}'f)(x) := \gamma x f'(x) + \frac{1}{2} x^2 \sigma(x)^2 f''(x), \quad \text{for } f \in C_0^2(\mathbb{E}).$$

By the Phillips theorem the infinitesimal generator $\mathcal{G}'_Z$ of the process $\{S'_{Z_t}\}_{t \geq 0}$ can be expressed as

$$(5.7) \quad \mathcal{G}'_Z = \psi_Z(-\mathcal{G}'),$$

where ψ_Z is the Laplace exponent of the Lévy subordinator Z . Formula (5.7) is a formal identity that can be interpreted in terms of functional calculus (see the references in the paragraph following Lemma 1 in Section 2). Representation (5.7) implies that the infinitesimal generator $\mathcal{G}$ of the process S is given by the following expression

$$(5.8) \quad (\mathcal{G}f)(x) = \mu x f'(x) + (\mathcal{G}'_Z f)(x) \quad \text{for any } f \in C_0^2(\mathbb{E}).$$

In order to construct the approximating Markov chain we fix a finite state-space $\mathbb{G} \subset \mathbb{E}$ (using the algorithm described in the beginning of this section) and define a generator matrix Λ' by the linear system in (5.2)–(5.4), the right-hand side of which is given by the drift coefficient γ and the diffusion coefficient σ of SDE (2.7) satisfied by the process S' . It is clear that the Markov chain X' , which corresponds to the generator Λ' , is by construction a Markov process that approximates the diffusion S' in the sense of Section 5.1.

We now define a generator matrix Λ'_Z by applying the Laplace exponent ψ_Z of the Lévy subordinator to the matrix $-\Lambda'$. Since the semigroup of the chain X' is generated by a bounded operator Λ' , the Phillips theorem implies that the matrix $\Lambda'_Z := \psi_Z(-\Lambda')$ is a generator matrix of the (time-changed) Markov chain $\{X'_{Z_t}\}_{t \geq 0}$ on the state-space $\mathbb{G}$. It is therefore natural to take the chain generated by Λ'_Z as an approximation for the process $\{S'_{Z_t}\}_{t \geq 0}$. Note that in order to compute the generator Λ'_Z we first obtain the decomposition $\Lambda' = UDU^{-1}$, where D is a diagonal matrix⁵ with eigenvalues equal to those of Λ' and U is an invertible matrix, and define $\Lambda'_Z := U\psi_Z(-D)U^{-1}$, where $\psi_Z(-D)$ is a diagonal matrix with eigenvalues $\psi_Z(-\lambda)$ when λ ranges over the spectrum of Λ' .

In order to obtain a matrix that approximates the operator given in (5.8), we have to add drift with state-dependent intensity to the Markov chain generated by the matrix Λ'_Z . We define a tri-diagonal generator matrix Λ_μ that satisfies the linear system

$$(5.9) \quad \sum_{y \in \mathbb{G}} \Lambda_\mu(x, y)(y - x) = (r - d)x - \sum_{z \in \mathbb{G}} \Lambda'_Z(x, z)(z - x) \quad \forall x \in \mathbb{G}^o$$

and $\Lambda_\mu(x, y) = 0$ for all $x \in \partial\mathbb{G}$, $y \in \mathbb{G}$. The generator matrix Λ that is used to approximate the operator in (5.8) can now be defined as

$$\Lambda := \Lambda'_Z + \Lambda_\mu.$$

This construction ensures that the chain X drifts locally at the same rate as the original process S . As in the diffusion case, the top and bottom rows of the generator Λ are set to be equal to zero.

⁵In practice the matrix Λ' always has a diagonal decomposition because the set of all square matrices that do not possess it is of codimension one in the space of all square matrices and therefore has Lebesgue measure zero.

5.3. Jump processes with state-dependent characteristics. In this section the asset price process S is assumed to be a non-negative Markov process with the generator in (2.11) with a possibly state-dependent volatility $\sigma(x)$ and jump measure $\nu(x, dy)$. The idea is to define the approximating Markov chain such that the first two instantaneous moments of the original process are matched, i.e. such that condition (5.1) holds. The construction of the generator matrix Λ of the approximating Markov chain X is now carried out in two steps: we first define the jump matrix Λ_J , which corresponds to the discretization of the jump measure ν , and then characterize a tri-diagonal generator matrix Λ_D by stipulating that the Markov chain X with the generator $\Lambda = \Lambda_J + \Lambda_D$ has the same instantaneous moments as the process S .

We start by building the state-space $\mathbb{G} \subset \mathbb{E}$ with N elements using the algorithm described in the beginning of this section. For any given $x \in \mathbb{G}^\circ$ we associate to $\mathbb{G}$ the set $\mathbb{G}_x \subset (-1, \infty)$ defined by

$$\mathbb{G}_x := \left\{ \frac{z}{x} - 1 : z \in \mathbb{G} \right\}.$$

The set $\mathbb{G}_x$ consists of the relative jump sizes of jumps starting from x and arriving at any other point in $\mathbb{G}$. The x -th row of the jump part Λ_J of the generator of X is obtained by discretizing the jump measure $\nu(x, dy)$ on the set $\mathbb{G}_x$. In particular let $\mathbb{G}_x = \{y_i : y_1 < \dots < y_{N-1}\}$ and define a function $\alpha_x : \mathbb{G}_x \rightarrow [-1, \infty]$ such that

$$(5.10) \quad \alpha_x(y_i) \in (y_i, y_{i+1}) \quad \text{for } i \in \{1, \dots, N-1\}$$

where $\alpha_x(y_0) = -1, \alpha_x(y_N) = \infty$. A possible natural choice for $\alpha_x(y_i)$ would be the mid-point of the interval

$$\alpha_x(y_i) = \frac{y_i + y_{i+1}}{2}.$$

We can now define the jump part of the generator as

$$(5.11) \quad \Lambda_J(x, x(1 + y_i)) := \int_{\alpha_x(y_{i-1})}^{\alpha_x(y_i)} \nu(x, dy)$$

where $i \in \{1, \dots, N\}$ and $y_i \neq 0$ and

$$(5.12) \quad \Lambda_J(x, x) := - \sum_{z \in \mathbb{G} \setminus \{x\}} \Lambda_J(x, z).$$

Note that the function α_x generates a partition of $(-1, \infty)$. The jump intensities of the chain, defined in formula (5.11), are obtained by integrating the Lévy measure over the corresponding part of the partition. For $x \in \partial\mathbb{G}$ we set $\Lambda_J(x, y) := 0$ for all $y \in \mathbb{G}$. It is clear that the matrix Λ_J constructed in this way is a generator matrix.

In the second step we match the first and second instantaneous moments of the asset price process S . In other words the chain X must satisfy condition (5.1) for all starting states $x \in \mathbb{G}^\circ$. Note that condition (5.1) implicitly assumes that the second instantaneous moment of S exists. This is the case if the jump measure satisfies the following condition

$$(5.13) \quad \int_{(-1, \infty)} y^2 \nu(x, dy) < \infty \quad \text{for all } x \in \mathbb{E}$$

which we now assume to hold.

The task now is to find a tri-diagonal generator matrix Λ_D such that the chain generated by the sum $\Lambda_D + \Lambda_J$ satisfies (5.1). The tri-diagonal matrix Λ_D therefore has to satisfy the following conditions

$$(5.14) \quad \sum_{z \in \mathbb{G}} \Lambda_D(x, z) = 0 \quad \text{and} \quad \Lambda_D(x, z) + \Lambda_J(x, z) \geq 0 \quad \forall z \in \mathbb{G} \setminus \{x\},$$

$$(5.15) \quad \sum_{z \in \mathbb{G}} \Lambda_D(x, z)(z - x) = (r - d)x - \sum_{z' \in \mathbb{G}} \Lambda_J(x, z')(z' - x),$$

$$(5.16) \quad \sum_{z \in \mathbb{G}} \Lambda_D(x, z)(z - x)^2 = x^2 \left[\sigma(x)^2 + \int_{-1}^{\infty} y^2 \nu(x, dy) \right] - \sum_{z' \in \mathbb{G}} \Lambda_J(x, z')(z' - x)^2$$

for all $x \in \mathbb{G}^o$, where r, d are the instantaneous interest rate and dividend yield respectively and σ is the local volatility function in (2.11). The right-hand side of equation (5.15) is the difference of the risk-neutral drift and the drift induced by the presence of jumps. Similarly the right-hand side of the linear equation in (5.16) consists of the difference of the instantaneous second moments of the asset price process S (computed directly from its generator (2.11)) and the chain that corresponds to the jump generator Λ_J . As usual we assume the absorbing boundary condition $\Lambda_D(x, y) = 0$ for all $x \in \partial\mathbb{G}$, $y \in \mathbb{G}$.

The linear system in (5.14)–(5.16) can typically be satisfied by a tri-diagonal generator matrix Λ_D if $\sigma(x)$ is strictly positive. Once we find Λ_D we define the generator matrix of the approximating chain X by

$$\Lambda := \Lambda_D + \Lambda_J.$$

6. CONVERGENCE AND ERROR ESTIMATES

6.1. Convergence of barrier option prices. Consider a sequence of finite-state continuous-time Markov chains $X^{(n)}$ approximating a given Feller price process $S = \{S_t\}_{t \geq 0}$. For $X^{(n)}$ to replicate as closely as possible the dynamics of S one chooses the generator matrix $\Lambda^{(n)}$ with the corresponding state-space $\mathbb{G}^{(n)}$ such that it is uniformly close to the infinitesimal generator $\mathcal{L}$ of S , in the sense that the distance $\epsilon_n(f)$ between the generators is small for a sufficiently large class $\tilde{\mathcal{D}}$ of regular test functions f , where

$$(6.1) \quad \epsilon_n(f) := \max_{x \in (\mathbb{G}^{(n)})^o} \left| \Lambda^{(n)} f_n(x) - \mathcal{L} f(x) \right|,$$

where $(\mathbb{G}^{(n)})^o$ equals $\mathbb{G}^{(n)}$ without the smallest and the largest elements and $f_n = f|_{(\mathbb{G}^{(n)})^o}$ is the restriction of f to $(\mathbb{G}^{(n)})^o$. More specifically, if $\epsilon_n(f)$ tends to zero as n tends to infinity for f in the class $\tilde{\mathcal{D}}$ and the probability that the chain exits $(\mathbb{G}^{(n)})^o$ before time T tends to zero, then the sequence of processes $(X^{(n)})_{n \in \mathbb{N}}$ converges weakly to the process S . This weak convergence on the level of the process implies in particular that the marginal distributions of $X^{(n)}$ will converge to those of S , and therefore the values of European options converge, that is,

$$\mathbf{E}_x[f(X_T^{(n)})] \rightarrow \mathbf{E}_x[f(S_T)]$$

for $x \in \mathbb{E}$, maturity $T > 0$, and continuous bounded functions f .

The payoff of the barrier option can be described in terms of the first-passage time of S and the position of S at that moment, which are both functionals of the path $\{S_t\}_{t \geq 0}$. For the weak convergence of $X^{(n)}$ to S to carry over to convergence of barrier-type payoffs, continuity (in the Skorokhod topology) is required of these two functionals, which is guaranteed to hold under Assumption 2. In view of the fact that the payoff of a barrier option is typically a discontinuous function, an additional condition is needed to ensure the convergence of the barrier option prices; we will assume that ℓ and u are such that

$$(6.2) \quad \mathbf{P}_x(S_T \in \{\ell, u\}) = 0.$$

Most models used in mathematical finance satisfy this condition. Even if (6.2) is not satisfied, this does not constitute a limitation in practice, since for any given process S the condition is satisfied for all but countably many pairs (ℓ, u) . The statement of the convergence is made precise in the following theorem, the proof of which is in Appendix A.3.

Theorem 2. *Let S be a Feller process with state-space $\mathbb{E}$ and infinitesimal generator $\mathcal{L}$ that does not vanish at zero and infinity.⁶ Let $(X^{(n)})_{n \in \mathbb{N}}$ be a sequence of Markov chains with generator matrices $\Lambda^{(n)}$ such that the rows corresponding to the smallest and the largest elements in $\mathbb{G}^{(n)}$ are equal to zero. Assume further that the following two conditions are satisfied for any function f in a core of $\mathcal{L}$:⁷*

$$(6.3) \quad \epsilon_n(f) \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

$$(6.4) \quad \text{either (i) } \lim_{y \searrow 0} \mathcal{L}f(y) = 0 \text{ or (ii) } \lim_{n \rightarrow \infty} \mathbb{P}_x \left[\tau_{(\mathbb{G}^{(n)})^o}^{(n)} > T \right] = 1,$$

where for any set $G \subset \mathbb{E}$ we define $\tau_G^{(n)} = \inf\{t \geq 0 : X_t^{(n)} \notin G\}$. If (6.2) holds, then, as $n \rightarrow \infty$,

$$\begin{aligned} \mathbf{E}_x \left[g \left(X_T^{(n)} \right) \mathbf{I}_{\{\tau_A^{(n)} > T\}} \right] &\longrightarrow \mathbf{E}_x \left[g(S_T) \mathbf{I}_{\{\tau_A > T\}} \right], \\ \mathbf{E}_x \left[e^{-r\tau_A^{(n)}} h \left(X_{\tau_A^{(n)}}^{(n)} \right) \mathbf{I}_{\{\tau_A^{(n)} \leq T\}} \right] &\longrightarrow \mathbf{E}_x \left[e^{-r\tau_A} h(S_{\tau_A}) \mathbf{I}_{\{\tau_A \leq T\}} \right], \end{aligned}$$

for any bounded continuous functions $g, h : \mathbb{E} \rightarrow \mathbb{R}$.

Remark 1. Since condition (6.3) is required to hold for all f in a core of $\mathcal{L}$, it follows that $\mathbb{G}^{(n)}$ will eventually ‘fill up’ the part of the state-space $\mathbb{E}$ where the generator does not vanish. Indeed, if there were to exist an open interval that does not intersect $\mathbb{G}^{(n)}$ for large n , then condition (6.3) would not hold for functions $f \in C^\infty(\mathbb{E})$ with compact support in this open interval satisfying $\mathcal{L}f \neq 0$, as $\Lambda^{(n)}f_n \equiv 0$. In particular, condition (6.3) implies that $x_1^{(n)} \rightarrow 0$ and $x_n^{(n)} \rightarrow \infty$ when $\mathbb{G}^{(n)} = \{x_1^{(n)}, \dots, x_n^{(n)}\}$ with $x_1^{(n)} < \dots < x_n^{(n)}$ if the generator does not vanish in a neighbourhood of zero and infinity.

⁶The set where the generator does not vanish is defined to be the set of all $x \in \mathbb{E} \cup \{\infty\}$ with the following property: for every open interval I in $\mathbb{E}$ that contains x (if $x = \infty$ then I takes the form (M, ∞) for some $M \in \mathbb{E}$) there exists a function $f \in \mathcal{D}$ with compact support in I such that the function $\mathcal{L}f$ is not identically equal to zero.

⁷A core $\mathcal{C}$ of the operator $\mathcal{L}$ is a subspace of the domain of $\mathcal{L}$ that is (i) dense in $C_0(\mathbb{E})$ and (ii) there exists $\lambda > 0$ such that the set $\{(\lambda - \mathcal{L})f : f \in \mathcal{C}\}$ is dense in $C_0(\mathbb{E})$.

Remark 2. Note that, since $\mathcal{L}f$ is by definition an element in $C_0(\mathbb{E})$ for any f in the domain of $\mathcal{L}$, it holds that $\mathcal{L}f(y) \rightarrow 0$ as $y \nearrow \infty$ and that $\mathcal{L}f(0)$ is well defined and equal to $\lim_{y \searrow 0} \mathcal{L}f(y)$. Assume that $\mathcal{L}$ does not vanish at zero and infinity and that the rows corresponding to the smallest and the largest elements in $\mathbb{G}^{(n)}$ of the generator matrices $\Lambda^{(n)}$ are equal to zero. Then, if conditions (6.3) and (6.4)(i) are satisfied, we have

$$(6.5) \quad \max_{x \in \mathbb{G}^{(n)}} \left| \Lambda^{(n)} f_n(x) - \mathcal{L}f(x) \right| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Remark 3. In practice the condition of boundedness of the payoff f is not restrictive as it is always possible to consider the truncation $f \wedge M$ for large constants M without losing noticeable accuracy. Further, under additional regularity properties on the parameters of the process S , the convergence in Theorem 2 also holds true for barrier call options. To see why this is the case note that

$$(6.6) \quad \mathbf{E}_x \left[(S_T - K)^+ \mathbf{I}_{\{\tau_A < T\}} \right] = x \bar{\mathbf{E}}_x \left[(1 - K/S_T)^+ \mathbf{I}_{\{\tau_A < T\}} \right]$$

where $\bar{\mathbf{E}}$ denotes the expectation under the measure $\bar{\mathbf{P}}$ given by $d\bar{\mathbf{P}}_x|_{\mathcal{F}_t} = x^{-1} S_t d\mathbf{P}_x|_{\mathcal{F}_t}$. Under $\bar{\mathbf{P}}$ the process S remains a Markov process, and, under additional regularity properties S is still a Feller process. The convergence then follows from Theorem 2 as the pay-off function on the right-hand side of (6.6) is bounded.

For the time-inhomogeneous case an analogous convergence result holds true. To approximate a given time-inhomogeneous Markov process $\{S_t\}_{t \in [0, T]}$, the sequence of time-inhomogeneous generator matrices $\Lambda^{(n)}$ of the form (4.7), defined on the time-space grids $[0, T] \times \mathbb{G}^{(n)}$, needs to be close to the time-dependent generator $\mathcal{L}_t$ of S (see Section 4 for the definition of $\mathcal{L}_t$), where the distance is measured by

$$\epsilon_n(f) := \max_{t \in [0, T], x \in \mathbb{G}^{(n)}} \left| \Lambda_t^{(n)} f_{t,n}(x) - \mathcal{L}_t f_t(x) \right|$$

for f in a core of $\mathcal{L}'$. Here f_t denotes the function $f_t : x \mapsto f(t, x)$ and $f_{t,n} = f_t|_{\mathbb{G}^{(n)}}$ the restriction of f_t to $\mathbb{G}^{(n)}$. Further, in the case of the rebate options a given short rate $r(t)$ also has to be approximated by an appropriately chosen piecewise constant function $r^{(n)}$ as in (4.8). Observe that the maximum in the definition above for ϵ_n runs over all of the points in $\mathbb{G}^{(n)}$, including the largest and the smallest elements. This differs from the definition for ϵ_n in (6.1) (cf. Remark 2).

Corollary 2. Let (D, Y) be the Feller process defined in (4.1) with state-space $\mathbb{E}' = [0, T] \times \mathbb{E}$ and infinitesimal generator $\mathcal{L}'$, and let $X^{(n)}$ be a sequence of time-inhomogeneous Markov chains with generator matrices $(\Lambda_t^{(n)}, t \in [0, T])$ such that the following conditions are satisfied:

$$(6.7) \quad \epsilon_n(f) \rightarrow 0 \quad \text{as } n \rightarrow \infty, \text{ for any function } f \text{ in a core of } \mathcal{L}',$$

$$(6.8) \quad \lim_{y \searrow 0} \mathcal{L}' f(t, y) = 0 \quad \text{for all } t \in [0, T].$$

If $r(t)$ is continuous and S satisfies (6.2) and Assumption 2 (with $\mathbf{P}_x$ replaced by $\mathbf{P}_{0,x}$), then, as $n \rightarrow \infty$,

$$\begin{aligned} \mathbf{E}_{0,x} \left[g \left(X_T^{(n)} \right) \mathbf{I}_{\{\tau_A^{(n)} > T\}} \right] &\longrightarrow \mathbf{E}_{0,x} \left[g(S_T) \mathbf{I}_{\{\tau_A > T\}} \right], \\ \mathbf{E}_{0,x} \left[e^{-\int_0^{\tau_A^{(n)}} r^{(n)}(t) dt} h \left(X_{\tau_A^{(n)}}^{(n)} \right) \mathbf{I}_{\{\tau_A^{(n)} \leq T\}} \right] &\longrightarrow \mathbf{E}_{0,x} \left[e^{-\int_0^{\tau_A} r(t) dt} h(S_{\tau_A}) \mathbf{I}_{\{\tau_A \leq T\}} \right], \end{aligned}$$

for any bounded continuous functions $g, h : \mathbb{E} \rightarrow \mathbb{R}$, where stopping times $\tau_A^{(n)}$ are as in Theorem 2.

6.2. Error estimates. In this section we quantify the speed at which the algorithm converges by providing error estimates for a specific choice of a sequence of approximating Markov chains, assuming sufficient smoothness of the value-function of the barrier option. Consider a Markov process S for which the corresponding time-space process (D, Y) forms a two-dimensional Feller process with infinitesimal generator acting on $f \in C^{1,2}([0, T] \times \mathbb{E})$ with compact support as

$$(6.9) \quad \mathcal{L}'v(t, x) = \frac{\partial v}{\partial t}(t, x) + \mathcal{L}_{D,t}v(t, x) + \mathcal{L}_{J,t}v(t, x) \quad \text{with}$$

$$(6.10) \quad \mathcal{L}_{D,t}v(t, x) := \frac{\sigma^2(t, x)x^2}{2}v''(t, x) + \gamma(t)xv'(t, x),$$

$$(6.11) \quad \mathcal{L}_{J,t}v(t, x) := \int_{(-1, \infty)} [v(t, x(1+y)) - v(t, x) - v'(t, x)xy]g(t, x, y)dy,$$

where $'$ denotes differentiation with respect to x , $\gamma(t) = r(t) - d(t)$ (r and d are the instantaneous interest rate and dividend yield respectively) and $g : [0, T] \times \mathbb{E} \times ((-1, \infty) \setminus \{0\}) \rightarrow \mathbb{R}_+$ is a nonnegative locally bounded function such that

$$\int_0^T \int_{(-1, \infty) \setminus \{0\}} g(s, x, y)(|y| \wedge y^2)dyds < \infty \quad \text{for all } x \in \mathbb{E},$$

where $x \wedge y = \min\{x, y\}$.

We next describe the sub-class of Markov processes that we will consider.

Definition 1. The process S is called *uniformly of bounded jump-variation* if the jump-density g satisfies the integrability condition

$$(6.12) \quad C_0 := \sup_{t \in [0, T], x \in [\ell, u]} \mathcal{I}(t, x) < \infty,$$

where

$$(6.13) \quad \mathcal{I}(t, x) := \int_{(-1, 1) \setminus \{0\}} |y|g(t, x, y)dy.$$

Definition 2. The process S is called *locally of stable type* on $[\ell, u]$ if there exist constants $\bar{\kappa}_\pm, \underline{\kappa}_\pm > 0$ and $\alpha_\pm \in (0, 2)$ such that for all $t \in [0, T]$ and $x \in [\ell, u]$

$$(6.14) \quad \frac{\underline{\kappa}_+}{|y|^{1+\alpha_+}} \leq g(t, x, y) \leq \frac{\bar{\kappa}_+}{|y|^{1+\alpha_+}}, \quad y \in (0, 1),$$

$$(6.15) \quad \frac{\underline{\kappa}_-}{|y|^{1+\alpha_-}} \leq g(t, x, y) \leq \frac{\bar{\kappa}_-}{|y|^{1+\alpha_-}}, \quad y \in (-1, 0).$$

We will consider three different cases:

- **Case O:** S is uniformly of bounded jump-variation.
- **Case I:** S is locally of stable type with $\alpha_- = 1$ or $\alpha_+ = 1$.
- **Case II:** S is locally of stable type with $\alpha_{\pm} \in (1, 2)$.

Case I can be considered to be a boundary case separating cases O and II, as S is uniformly of bounded jump-variation if it is of stable type with $\alpha_{\pm} < 1$. The class of processes of uniformly bounded jump-variation (case O) contains processes S whose jump-part forms a compound Poisson process (such as the Kou model [39] and the CEV-Kou model, described in Section 7.3), as well as the Lévy models of bounded variation such as for example the VG model. Examples of processes satisfying the condition in cases I and II are the Generalised Hyperbolic Lévy models and CGMY models with $Y > 1$, respectively. More generally, the class of processes that are locally of stable type contains the class of *regular Lévy processes of exponential type* studied in Boyarchenko and Levendorskii [7].

We will impose in addition the following regularity conditions:

Assumption 3. *There exists a constant C such that for all $s, t \in [0, T]$ and $x \in [\ell, u]$*

$$|r(t) - r(s)| + |d(t) - d(s)| + |\sigma^2(t, x) - \sigma^2(s, x)| \\ + \int_{(-1, \infty) \setminus \{0\}} |g(t, x, y) - g(s, x, y)| [y^2 \wedge |y|] dy \leq C|t - s|.$$

Assumption 4. *There exists a constant C' such that for all $t \in [0, T]$ and $x \in [\ell, u]$*

$$\sigma^2(t, x)x^2 + |\gamma(t)|x + \int_{(-1, \infty) \setminus \{0\}} (y^2 \wedge |y|)g(t, x, y)dy \leq C'.$$

Assumption 5. *The pay-off function f is Lipschitz-continuous⁸ and has compact support. The barrier option value-function $\tilde{Q}_t^A f(0, x) = \mathbb{E}_{0,x}[e^{-\int_0^{t \wedge \tau_A} r(s)ds} f(S_{t \wedge \tau_A})]$ is $C^{1,3}([0, T] \times \mathbb{E})$.*

Remark 4. In the strictly elliptic case ($\sigma(t, x) > 0$ for $(t, x) \in [0, T] \times [\ell, u]$) with a jump-part that is uniformly of bounded variation, explicit sufficient conditions on the data f, g and σ^2 guaranteeing the smoothness of $v_A : (t, x) \mapsto \tilde{P}_t^A f(x)$ required in Assumption 5 follow from classical existence and uniqueness results for Cauchy-Dirichlet problems associated to second order partial-integro differential operators. Specifically, Theorem II.3.3 in Garroni & Menaldi [24] implies that $v_A \in C^{1,3}([0, T] \times [\ell, u])$ if, for some $0 < \epsilon < 1$, the following conditions are satisfied: (a) f has compact support in (ℓ, u) and $f \in C^{2+\epsilon}([\ell, u])$ ⁹, (b) $\sigma^2 \in C^{1,1}([0, t] \times [\ell, u])$, (c) $x \mapsto g(t, x, y)|y|^{2+\epsilon}$ is $C^1([\ell, u])$ with bounded derivatives, uniformly over $(t, x, y) \in [0, T] \times [\ell, u] \times (-1, \infty)$ and (d) r and d are continuous on $[0, T]$. Note that (c) implies that the $\mathcal{L}_J v$ in (6.11) is $C^{1,1}([0, T] \times [\ell, u])$ if v is $C^{1,2}([0, T] \times [\ell, u])$.

Throughout the rest of this section we take

$$0 < \ell < u < \infty$$

⁸A function $f : \mathbb{E} \rightarrow \mathbb{R}$ is Lipschitz continuous if there exists a constant $L > 0$ such that $|f(x) - f(y)| \leq L|x - y|$ for all $x, y \in \mathbb{E}$.

⁹ $C^{n+\epsilon}([\ell, u])$ is the set of functions $f : [\ell, u] \rightarrow \mathbb{R}$ that are n times continuously differentiable with the n th derivative ϵ -Hölder continuous.

and consider the spatial and time grids given by

$$(6.16) \quad \begin{aligned} \mathbb{G}^{(n)} &= \{x_i^{(n)} : 0 = x_1^{(n)} < \dots < x_n^{(n)}\}, \\ \mathbb{T}^{(m)} &= \{T_j^{(m)} : 0 = T_0^{(m)} < \dots < T_m^{(m)} = T\}. \end{aligned}$$

We will denote by

$$(6.17) \quad h(n) = \max_i |x_{i+1}^{(n)} - x_i^{(n)}|, \quad \delta(m) = \max_j |T_{j+1}^{(m)} - T_j^{(m)}|,$$

the meshes of the spatial and time grids, and by

$$(6.18) \quad k(n) = \max_{x_k^{(n)} \in \widehat{\mathbb{G}}^{(n)}} \sup_{t \in [0, T]} \int_{(-1, \infty) \setminus [L(x_k^{(n)}), U(x_k^{(n)})]} g(t, x_k^{(n)}, y) dy,$$

the tail mass of the jump-measure, where $\widehat{\mathbb{G}}^{(n)} = \mathbb{G}^{(n)} \cap [\ell, u]$ and

$$L(x_k^{(n)}) := \frac{x_2^{(n)}}{x_k^{(n)}} - 1 \quad \text{and} \quad U(x_k^{(n)}) := \frac{x_{n-1}^{(n)}}{x_k^{(n)}} - 1$$

are the relative jump-sizes from $x_k^{(n)}$ to the one-but-largest and one-but-smallest elements $x_2^{(n)}$ and $x_{n-1}^{(n)}$ of the state-space $\mathbb{G}^{(n)}$. We approximate S by a sequence of time-inhomogeneous Markov chains $(X^{(n,m)})_{n,m \in \mathbb{N}}$ with infinitesimal generators acting on functions $g : [0, T] \times \mathbb{G}^{(n)} \rightarrow \mathbb{R}$ as

$$\frac{\partial g}{\partial t} + \Lambda_t^{(n,m)} g,$$

where $\Lambda_t^{(n,m)}$ is piecewise constant in t , given by

$$(6.19) \quad \Lambda_t^{(n,m)} = \sum_{j=1}^m \Lambda^{(j)} \mathbf{1}_{[T_{j-1}, T_j)}(t).$$

The matrices $\Lambda^{(j)}$, $j = 1, \dots, m$, are given as $\Lambda^{(j)} = \Lambda_D^{(j)} + \Lambda_J^{(j)}$ for some generator matrices $\Lambda_D^{(j)}$ and $\Lambda_J^{(j)}$ which approximate the infinitesimal generators of the diffusion and purely discontinuous parts of S at time t_j . An explicit description of $\Lambda_D^{(j)}$ and $\Lambda_J^{(j)}$ is given in Appendix A.5.

Theorem 3. *Let Assumptions 3, 4 and 5 hold, and consider cases O, I or II. Assume also that the sequences of grids $\mathbb{G}^{(n)}$ and $\mathbb{T}^{(m)}$ satisfy*

$$\lim_{n \rightarrow \infty} h(n) = \lim_{n \rightarrow \infty} k(n) = \lim_{m \rightarrow \infty} \delta(m) = 0.$$

Then there exist constants C_1, C_2, C_3 , independent of $h(n)$, $k(n)$ and $\delta(m)$, given in (6.17) and (6.18), such that for all n and m sufficiently large and $x \in \mathbb{G}^{(n)}$

$$\left| \mathbf{E}_{0,x} \left[e^{-\int_0^{T \wedge \tau_A^{(n)}} r^{(n)}(t) dt} f \left(X_{T \wedge \tau_A^{(n)}}^{(n)} \right) \right] - \mathbf{E}_{0,x} \left[e^{-\int_0^{T \wedge \tau_A} r(t) dt} f(S_{T \wedge \tau_A}) \right] \right| \leq C_1 E(h) + C_2 k + C_3 \delta,$$

where $h = h(n)$, $\delta = \delta(m)$ and $k = k(n)$ and

$$(6.20) \quad E(h) = \begin{cases} h, & \text{in cases O and II,} \\ -h \log h, & \text{in case I.} \end{cases}$$

$\sigma_0 = 0.2, r = 0.02$ $K = 2, \ell = 1.5, u = 2.5$			$\sigma_0 = 0.5, r = 0.05$ $K = 2, \ell = 1.5, u = 3$			$\sigma_0 = 0.5, r = 0.05$ $K = 1.75, \ell = 1, u = 3$		
GY	KI	MG	GY	KI	MG	GY	KI	MG
0.0411	0.041089	0.041082	0.0178	0.017856	0.017856	0.07615	0.076172	0.076165

TABLE 1. The comparison of double barrier call option prices with strike K obtained in [26] and [41] in the case of geometric Brownian motion. The model is given by SDE (2.7) with the constant volatility function $\sigma(x) := \sigma_0$ and drift $\gamma = r - d$, where the interest rate r is given in the table and the dividend yield equals $d = 0$. The asset price process S starts at $S_0 = 2$ and the maturity in all the cases is $T = 1$ year. The state-space of the approximating chain is defined by the algorithm in Section 5 and the parameters $N = 200$, $x_1 = 0.2$, $x_N = 10$, $d_1^- = 100$, $d_1^+ = 1$, $d_2^- = 10$, $d_2^+ = 10$, $d_3^- = 1$, $d_3^+ = 100$. The computation for the pricing of the barrier products takes about 0.03 seconds (using Matlab(R) 7.8 on Intel(R) Xeon(R) CPU E5430 @ 2.66GHz) for each of the parameter choices in this table.

The proof of Theorem 3 is given in Appendix A.6. The logarithmic factor in the error bound (6.20) in case I is due to the form (6.14)–(6.15) of the singularity of the jump measure at zero in this case (see the proof for details). Note that the discontinuous part of a process satisfying case I is of infinite variation, but has zero $(1 + \epsilon)$ -variation for every $\epsilon > 0$.

Numerical experiments, reported in Section 7, suggest that in several pricing models of interest the error of the Markov generator (MG) algorithm described in Section 5 is actually of order h^2 . A further theoretical investigation of error bounds and sharp rates of convergence of the algorithm, is left for future research.

7. NUMERICAL EXAMPLES

In this section the Markov chain algorithm is examined numerically in a variety of contexts. Subsection 7.1 contains numerical examples for diffusion based models and a comparison with alternative methods for the pricing of barrier options in this setting (binomial trees and the finite difference Crank-Nicholson algorithms). In subsections 7.2 and 7.3 the algorithm is applied to the cases of Lévy driven SDEs and jump-diffusion models with time-dependent characteristics.

7.1. Diffusions.

7.1.1. Geometric Brownian motion. The model is given by SDE (2.7) where the volatility function $\sigma(x) = \sigma_0$ is constant and the drift equals $\gamma = r - d$, where r is the risk free rate and d is the dividend yield. We now compare our algorithm (MG), based on the Markov generator of the approximating chain X , with the results obtained in Geman and Yor [26] and Kunitomo and Ikeda [41]. The numerical results are contained in Table 1.

In Figure 3 the errors and computation times of the Markov generator algorithm (MG) and a standard binomial tree are plotted against the number of points, for the computation of the price of a double knock-out barrier call option. The true value of this option (0.0410886) with parameters given in the first section in Table 1 was obtained using the eigenfunction expansion for the double-barrier option prices in the Black-Scholes model given in [18]). Note that the graph of

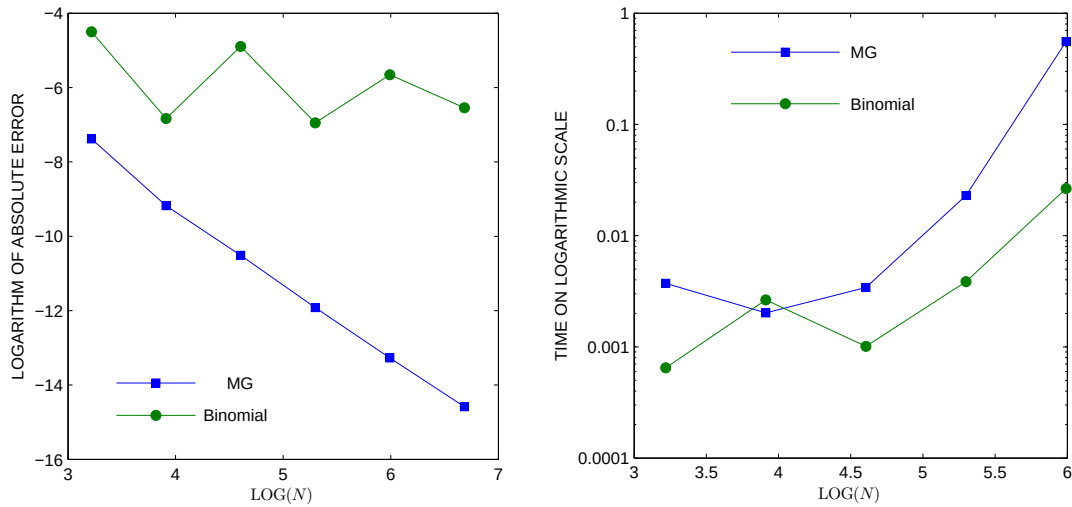


FIGURE 3. The first figure depicts the pricing errors of the generator method (MG) and the binomial tree algorithm for the pricing of barrier options. The barrier contract and the model parameters of the underlying geometric Brownian motion are given in the first section in Table 1. The true value of this contract used to compute the pricing errors equals 0.0410886 (which was obtained by the eigenfunction expansion in [18]). The first figure is on a “log-log” scale where the vertical axes contains the logarithm of the absolute value of the error and the horizontal axis depicts the logarithm of the number of states N in the grid. The second figure shows the computation time (in seconds) of the two algorithms as a function of $\log(N)$. Note that in the case of MG algorithm, the integer N equals the actual number of states in the grid while in the binomial tree method the errors and computation times that corresponds to $\log(N)$ arise in the model with $2N - 1$ states and time increment of the size T/N , where $T = 1$ is the maturity of the option.

the logarithm of the absolute pricing error for the MG algorithm given in Figure 3 is approximately linear in $\log(N)$ with slope -2 , which implies that the error itself is approximately quadratic in $1/N$. The well known jagged pattern phenomenon in the error of the pricing algorithm based on the binomial tree can also be observed in Figure 3. This is a consequence of the fact that the barrier levels do not always coincide with nodes of the binomial tree. Such behaviour is not present in the case of the MG algorithm because it employs a non-uniform grid that always contains the barrier levels (see the beginning of Section 5 for a description of the construction of the grid).

Figure 3 shows the well-known zig-zag behaviour of the standard binomial tree algorithm. In the case of geometric Brownian motion, the pricing algorithm based on a discrete-time Markov chain can be improved by regularizing the payoff function and choosing the state-space of the approximating discrete-time chain so that it contains the barriers and the strike of the derivative that is being priced. Such improvements, using trinomial trees, have been proposed in Ritchken [54]. We now compare the MG algorithm with the numerical results of [54].

In Figure 4 the errors and computation times of the Markov generator algorithm and a trinomial tree are plotted against the number of points, for the computation of the price of a double knock-out barrier call option. Note that the graph of the logarithm of the absolute pricing error for the MG algorithm given in Figure 4 is, as in Figure 3, approximately linear in $\log(N)$ with slope -2 , which

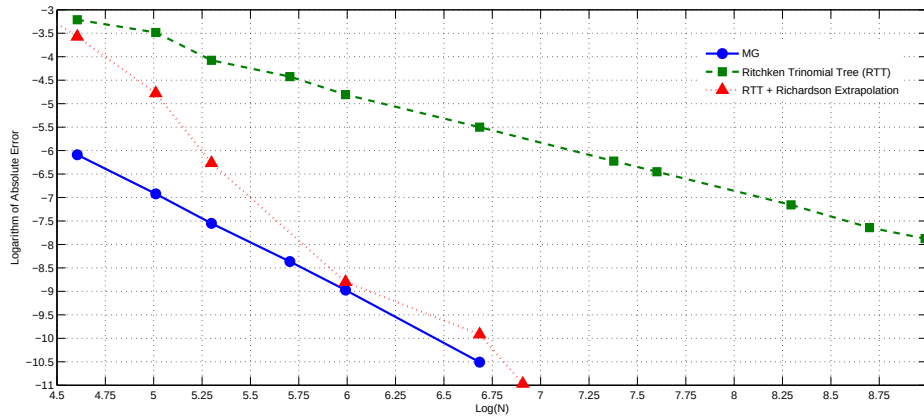


FIGURE 4. This figure depicts the pricing errors for the double barrier knock-out call option ($S_0 = 95$, $K = 100$, $\ell = 90$, $u = 140$, $T = 1$, $r = 0.1$, $\sigma_0 = 0.25$) of the MG algorithm, the trinomial tree method described in Ritchken [54, Ex. 7] and its second order Richardson extrapolation. The true value of this contract used to compute the pricing errors equals 1.4583798 (this was obtained by the MG algorithm with $N = 3000$ states). The figure is on a “log-log” scale where the vertical axes contains the logarithm of the absolute value of the error and the horizontal axis depicts the logarithm of the number of states N in the grid.

as before implies that the error itself is approximately quadratic in $1/N$. The pricing algorithm, given in Ritchken [54, Ex. 7], is based on the trinomial tree and it appears to converge at the rate $1/N$. Figure 4 shows that when the trinomial tree algorithm of Ritchken [54, Ex. 7] is combined with the second order Richardson extrapolation, the convergence appears to be of higher order than in the MG algorithm.

7.1.2. CEV process. The model is given by SDE (2.7) where the volatility function takes the form $\sigma(x) := \sigma_0(x/S_0)^\beta$ (S_0 is the starting value of the process) and the drift equals $\gamma = r - d$ where r is the risk free rate and d denotes the dividend yield. Table 2 contains the numerical results of the algorithm (MG) as well as the results obtained in Davydov and Linetsky [17]. See also Appendix B for the implementation of the pricing algorithm in Matlab that produced these results. To download the code see [51].

In Figure 5 the errors and computational times of both the MG algorithm and the finite difference Crank-Nicholson PDE scheme on a uniform grid (reported in [18]) are plotted against the number of points in the state-space, for the case of a barrier option driven under the CEV model. As an example we take a double barrier knock-out call option with expiry $T = 1$ year, strike $K = 100$ and lower and upper barriers given by $\ell = 80$, $u = 120$ respectively. The true price of this contract under the CEV model with parameter values given in Table 2 (for $\beta = -4$) is given by 0.5822268 (see [18], Table 7, where the eigenfunction expansion is used to calculate the exact price). The figure shows that the error of the Markov generator algorithm again decreases quadratically in $1/N$.

7.2. Lévy and local Lévy models.

CEV			$\beta = 0$			$\beta = -0.5$			$\beta = -1$		
u	ℓ	K	MG: $n = 2$	MG: $n = 3$	DL	MG: $n = 2$	MG: $n = 3$	DL	MG: $n = 2$	MG: $n = 3$	DL
120	90	95	1.7038	1.7038	1.7039	1.8805	1.8805	1.8805	2.0799	2.0800	2.0800
120	90	100	0.9703	0.9703	0.9703	1.0957	1.0957	1.0958	1.2382	1.2383	1.2383
120	90	105	0.4417	0.4418	0.4418	0.5124	0.5125	0.5126	0.5943	0.5944	0.5945
			$\beta = -2$			$\beta = -3$			$\beta = -4$		
u	ℓ	K	MG: $n = 2$	MG: $n = 3$	DL	MG: $n = 2$	MG: $n = 3$	DL	MG: $n = 2$	MG: $n = 3$	DL
120	90	95	2.5527	2.5528	2.5529	3.1292	3.1294	3.1295	3.8084	3.8087	3.8088
120	90	100	1.5797	1.5798	1.5799	2.0019	2.0021	2.0022	2.5055	2.5058	2.5059
120	90	105	0.7958	0.7960	0.7960	1.0532	1.0534	1.0535	1.3693	1.3696	1.3696

TABLE 2. The comparison of double barrier call option prices with strike K obtained in [17] using the spectral decomposition of the CEV process and our algorithm based on the Markov generator. The model is given by SDE (2.7) with the volatility function $\sigma(x) := \sigma_0(x/S_0)^\beta$ and drift $\gamma = r - d$, where r is the interest rate and d is the dividend yield. The model parameters are as follows: $S_0 = 100$, $\sigma_0 = 0.25$, $r = 0.1$, $d = 0$ and all options in the table expire at maturity $T = 0.5$. The state-space of the approximating chain is defined by the algorithm in Section 5 and the parameters $N = 2^n \cdot 100$ (where $n = 2, 3$), $x_1 = 0.1, x_N = 190$, $d_1^- = 100$, $d_1^+ = 1$, $d_2^- = 10$, $d_2^+ = 10$, $d_3^- = 1$, $d_3^+ = 100$. The computation time for the pricing of all the barrier options takes about one second in the case $n = 2$ and about ten seconds when $n = 3$ on the same hardware as in Table 1 for each of the six models in this table.

7.2.1. *The CGMY/KoBoL process.* In this section we assume that the price S is again an exponential Lévy process given by (2.9), where the Lévy processes L is a CGMY process [10] with Lévy density given by the formula

$$(7.1) \quad k(y) := C \left(\mathbf{I}_{(-\infty, 0)} \frac{e^{-G|y|}}{|y|^{Y+1}} + \mathbf{I}_{(0, \infty)} \frac{e^{-My}}{|y|^{Y+1}} \right), \quad \text{where } M > 1, G \geq 0, C > 0, Y < 2.$$

The inequality $Y < 2$ is induced by the integrability condition on the Lévy measure at zero and the condition $M > 1$ implies the exponential moment condition (2.10).

Madan and Yor [50] show that there exists a Lévy subordinator Z with the Laplace exponent ψ_Z given by

$$(7.2) \quad \mathbf{E}_0[e^{-uZ_t}] = e^{t\psi_Z(u)} = \exp(t\Gamma(-Y)[2r(u)^Y \cos(\eta(u)Y) - M^Y - G^Y]), \quad u \geq -\frac{GM}{2},$$

where Γ denotes the Gamma function and the functions r, η are given by the formulae

$$r(u) := \sqrt{2u + GM} \quad \text{and} \quad \eta(u) := \arctan\left(\frac{2\sqrt{2u - \theta^2}}{G + M}\right).$$

Note that for $u \in [-GM/2, \theta^2/2)$ the function η in formula (7.2) takes purely imaginary values which are mapped by the cosine function into the real numbers. Furthermore it is shown in [50] that the Lévy process L , given by the Lévy density (7.1), has the same law as the process $\{W_{Z_t} + \theta Z_t\}_{t \geq 0}$ where $\theta := (G - M)/2$ and the Brownian motion W is independent of the subordinator Z . Therefore we have the identity $\mathbf{E}_0[\exp(L_1)] = \exp(\psi_Z(-\theta - 1/2))$ which (together with definition (2.9)) implies that the process S has the same law as the process given by (5.5) where S' is a geometric Brownian motion and the drift μ satisfies $\mu = r - d - \psi_Z(-\theta - 1/2)$ by (5.6). We can therefore apply

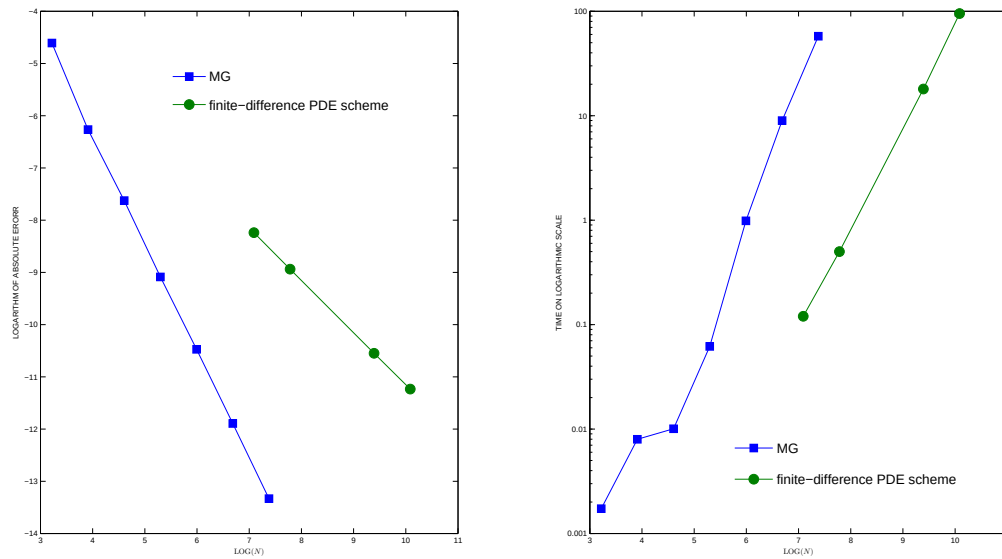


FIGURE 5. The first figure displays the pricing errors for the MG algorithm and the finite difference Crank-Nicholson PDE scheme for the double barrier call option with expiry $T = 1$ year, strike $K = 100$ and lower and upper barriers given by $\ell = 80$, $u = 120$ respectively in the CEV model with $\beta = -4$ and other parameter values given in the caption of Table 2. The exact price of this contract is given by 0.5822268 and the errors and the computation times for a standard finite difference Crank-Nicholson scheme are based on [18], Table 7. The first figure is on a “log-log” scale where the vertical axes contains the logarithm of the absolute value of the error and the horizontal axis depicts the logarithm of the number of states N in the grids for the MG algorithm and the PDE solver. The second figure shows the computation time (in seconds) of the two algorithms as a function of $\log(N)$.

the algorithm described in Section 5.2. We compare our numerical results with those obtained in Boyarchenko and Levendorskii [6] using a Fourier method (see Table 3).

In Figure 6 the error and computation times are plotted of the prices of double-no-touch and double knock-out put options as a function of the grid size. As the true values we took the outcomes of the algorithm for $N = 6400$. The figure appears to suggest that the error is approximately proportional to $N^{-1.2}$ and N^{-2} , respectively, where N is the number of point in the grid. Figure 6 depicts the prices as a function of the spot value, with the values given in Table 3. Close to the barrier levels the price the delta (derivative with respect to spot) appears to be infinite. The behaviour of the option prices near the barrier levels for Lévy models is studied theoretically in Levendorskii [44].

7.2.2. GH process. The price process S in this example is assumed to be an exponential Lévy process given by (2.9) where L is a *generalised hyperbolic* (GH) Lévy process with distribution at

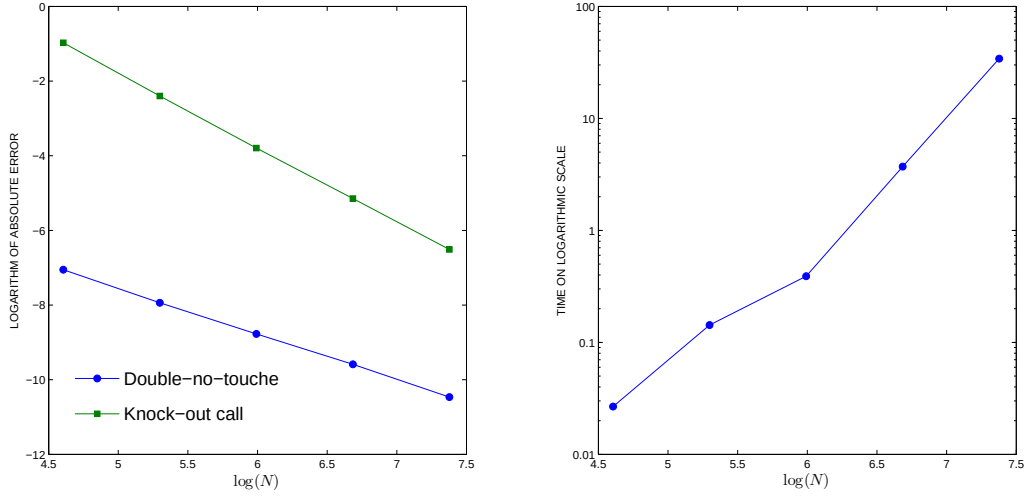


FIGURE 6. The first figure displays the logarithm of the absolute pricing errors for the MG algorithm applied to the CGMY model as a function of $\log(N)$, where N is the number of states in the grid. Under consideration are the double no touch and double knock out call options with $K = S_0 = 3500$, lower barrier $\ell = 2800$ and upper barrier $u = 4200$, driven by a CGMY process with parameters as in Table 3. As true values we took the values 78.752 and 0.9508 obtained by running the algorithm with $N = 6400$ points. The second figure shows the computation time (in seconds) of the algorithm as a function of $\log(N)$.

Spot	Knock-out put		Double-no touch	
%	BL	MG	BL	MG
82	302.28	301.07	0.5778	0.5757
85	370.38	370.38	0.8009	0.8004
88	341.35	341.78	0.8881	0.8880
91	279.86	280.41	0.9280	0.9280
94	207.71	208.30	0.9464	0.9465
97	136.63	137.24	0.9527	0.9529
100	78.19	78.74	0.9506	0.9507

Spot	Knock-out put		Double-no touch	
%	BL	MG	BL	MG
101	64.07	64.53	0.9481	0.9483
104	36.96	37.18	0.9351	0.9352
107	22.73	22.84	0.9112	0.9113
110	14.60	14.65	0.8708	0.8709
113	9.61	9.64	0.8020	0.8019
116	6.30	6.32	0.6771	0.6767
119	3.52	3.54	0.4049	0.4049

TABLE 3. Barrier option prices under the CGMY model. The first column contains the spot price as percentage of 3500. The CGMY parameters are $C = 1$, $G = 9$, $M = 8$, $Y = 0.5$. The resulting risk-neutral drift is $\mu = r - d - \psi_Z(-\theta - 1/2) \approx -0.0423$. Option parameters $K = 3500$ (strike of the put), $\ell = 2800$, $u = 4200$, $r = 0.03$, $d = 0$, $T = 0.1$. The columns BL and MG report the results obtained by Boyarchenko and Levendorskii [6] and by the Markov generator algorithm (with $N = 800$ points) respectively. It takes about 22 seconds to run the MG algorithm for each starting spot price on the same hardware as in Table 1.

time 1 given by the characteristic function

$$(7.3) \quad \Phi_{L_1}(u) := \mathbf{E}_0[e^{iuL_1}] = e^{imu} \left(\frac{\alpha^2 - \beta^2}{\alpha^2 - (\beta + iu)^2} \right)^{\lambda/2} \frac{K_\lambda(\delta\sqrt{\alpha^2 - (\beta + iu)^2})}{K_\lambda(\delta\sqrt{\alpha^2 - \beta^2})}$$

where $m, \lambda \in \mathbb{R}$, $\alpha, \delta > 0$, $0 \leq |\beta| < \alpha$.

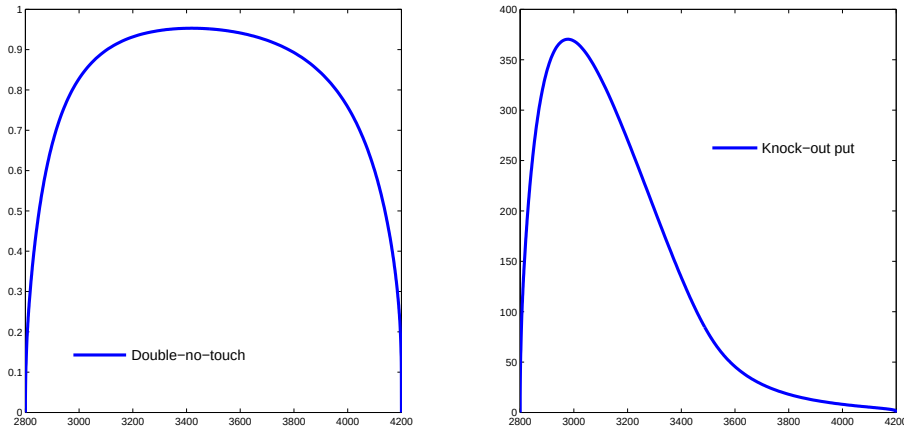


FIGURE 7. Plotted are prices of the double-no-touch and double knock out put options for different values of the spot S_0 with lower barrier $\ell = 2800$, upper barrier $u = 4200$ and strike $K = 3500$, driven by a CGMY process with parameters as in Table 3.

The function K_λ is known as the modified Bessel function of the second kind (see [1] for the precise definition of K_λ). The corresponding distribution is called *generalised hyperbolic* and is denoted by $\text{GH}(\lambda, \alpha, \beta, \delta, m)$. The class of distributions described by the characteristic function in (7.3) was introduced into the mathematical finance literature by Eberlein and Keller [22] and has been studied extensively ever since (see e.g. Prause [53] and the references therein). Since L has to satisfy the exponential moment condition (2.10), we have to restrict the parameter space in the following way (see for example Lemma 1.13 in [53] for details):

$$(7.4) \quad m, \lambda \in \mathbb{R}, \quad \alpha, \delta > 0 \quad \text{and} \quad \left| \beta + \frac{1}{2} \right| + \frac{1}{2} < \alpha.$$

Furthermore it is clear from (2.9) that the value of the parameter m in (7.3) has no bearing on the model S and can without loss of generality be taken equal to zero.

Let Z be a *generalised inverse Gaussian* (GIG) subordinator with the Laplace exponent ψ_Z given by

$$(7.5) \quad \mathbf{E}_0 [e^{-uZ_1}] = e^{\psi_Z(u)} = \left(\frac{a}{a+2u} \right)^{\lambda/2} \frac{K_\lambda \left(\sqrt{b(a+2u)} \right)}{K_\lambda \left(\sqrt{ab} \right)}, \quad \text{where } \lambda \in \mathbb{R}, a, b > 0.$$

The distribution of Z_1 is called *generalised inverse Gaussian* and denoted by $\text{GIG}(\lambda, a, b)$. It follows from expressions (7.3) and (7.5) that if we choose $Z_1 \sim \text{GIG}(\lambda, \alpha^2 - \beta^2, \delta^2)$ where the parameters $\lambda, \alpha, \beta, \delta$ satisfy condition (7.4), then the equality

$$\mathbf{E}_0 [e^{iuL_1}] = \exp \left(\psi_Z \left(u^2/2 - i\beta u \right) \right)$$

holds for all $u \in \mathbb{R}$. Therefore L has the same law as the process $\{\beta Z_t + W_{Z_t}\}_{t \geq 0}$ where W is a Brownian motion independent of Z . Hence the asset price process S obviously possesses the structure in (5.5) where the diffusion S' is a geometric Brownian motion and the drift equals

Generalised hyperbolic Lévy process $\alpha = 26.4, \beta = -0.53, \delta = 0.0034, \lambda = -0.5$	$n = 1$	$n = 2$	$n = 3$	$n = 4$	$n = 5$
KO Call: $K = 100, \ell = 97, u = 105$	1.0193	1.0190	1.0188	1.0187	1.0187
Double-no-touch: $\ell = 97, u = 105$	0.9704	0.9711	0.9714	0.9716	0.9716

TABLE 4. The prices of the double barrier knock-out call option and the double-no-touch option in the generalised hyperbolic Lévy model. The parameter values for the Lévy process L are taken from [53], page 64, Table 2.27, and are given in the table. The strike in the case of the call option is given by $K = 100$ and the barrier levels for both derivatives are equal to $\ell = 97, u = 105$. We take $S_0 = 100$, $r = 0.01$, $d = 0$ and $T = 1$. The risk-neutral drift μ in (5.5) takes the form $\mu = r - d - \psi_Z(-\beta - 1/2) \approx -0.0099$. The state-space of the approximating chain is defined by the algorithm in Section 5 and the parameters $N = 2^n \cdot 100$ (where $n = 2, \dots, 5$), $\min S = 50$, $\max S = 150$, $g_l^L = 100$, $g_u^L = 1$, $g_l^S = 10$, $g_u^S = 10$, $g_l^U = 1$, $g_u^U = 100$. The approximate computation time of the pricing algorithm (on the same hardware as in Table 1) for the case $n = 3$ is 19 seconds.

$\mu = r - d - \psi_Z(-\beta - 1/2)$. We can therefore apply the algorithm described in Section 5.2. Table 4 contains the numerical results.

7.2.3. *A local Lévy model.* A Markov process S with state-dependent characteristics, which starts at $S_0 > 0$ and has double-exponential jumps, can be specified by the following Lévy driven SDE

$$(7.6) \quad \frac{dS_t}{S_{t-}} = (r - d - \lambda \zeta(S_{t-}/S_0)^\beta) dt + (S_{t-}/S_0)^\beta dL_t, \quad \text{where}$$

$$(7.7) \quad L_t := \sigma_0 W_t + \sum_{i=1}^{N_t} (e^{K_i} - 1), \quad \sigma_0 \in (0, \infty) \quad \text{and} \quad \beta \in \mathbb{R}.$$

The special case of this model for $\beta = 0$ was introduced into the mathematical finance literature by Kou [39]. The random variables K_i , $i \in \mathbb{N}$, are independent of both the Brownian motion W and the Poisson process N with intensity $\lambda > 0$ and are distributed according to the double exponential density

$$(7.8) \quad f_K(k) = p\eta_1 e^{-\eta_1 k} \mathbf{I}_{(0, \infty)} + (1 - p)\eta_2 e^{\eta_2 k} \mathbf{I}_{(-\infty, 0)}, \quad \text{where} \\ \eta_1 > 1, \eta_2 > 0 \quad \text{and} \quad p \in [0, 1].$$

The parameter ζ is given by

$$\zeta := \mathbf{E}[e^{K_1} - 1] = \frac{p\eta_1}{\eta_1 - 1} + \frac{(1 - p)\eta_2}{\eta_2 + 1} - 1.$$

If $\beta < 0$ the process S has positive probability of hitting zero in finite time, in which case we take 0 to be absorbing.

It is clear that the model described by (7.6) and (7.7) has a generator of the form given in (2.11) with $\sigma(x) = \sigma_0(x/S_0)^\beta$ and $\mu(x) = \lambda \zeta(x/S_0)^\beta$. The jump measure $\nu(x, dy)$ in representation (2.11) is supported in $(-1, \infty)$ and in our case by (7.8) takes the explicit form

$$(7.9) \quad \nu(x, dy) = (x/S_0)^\beta \lambda [p\eta_1(y + 1)^{-1-\eta_1} \mathbf{I}_{(0, \infty)} + (1 - p)\eta_2(y + 1)^{\eta_2-1} \mathbf{I}_{(-1, 0)}] dy.$$

Local Lévy model		MG: $N = 400$	MG: $N = 800$	MG: $N = 1200$	KW
$\beta = 0$	$\lambda = 3$	10.0528	10.0530	10.0530	10.05307
	$\lambda = 0.01$	9.2768	9.2771	9.2772	9.27724
$\beta = -1$	$\lambda = 3$	9.7685	9.7688	9.7688	N/A
	$\lambda = 0.01$	8.9572	8.9575	8.9575	N/A
$\beta = -3$	$\lambda = 3$	9.0185	9.0187	9.0188	N/A
	$\lambda = 0.01$	8.0855	8.0858	8.0858	N/A

TABLE 5. Up-and-in call option prices under the model S defined in (7.6) and (7.7). The parameters are given by $S_0 = 100$, $r = 5\%$, $d = 0$, $\sigma_0 = 0.2$, $p = 0.3$, $1/\eta_1 = 0.02$, $1/\eta_2 = 0.04$, $\beta \in \{0, -1, -3\}$ and $\lambda \in \{0.01, 3\}$. The strike is $K = 100$ and the upper barrier $u = 120$ while time to maturity $T = 1$. The column KW denotes the results of Kou and Wang (see [40], Table 3) while MG denotes the algorithm based on the Markov generator. The state-space of the approximating chain is defined by the algorithm analogous to the one in Section 5, adapted in an obvious way to a single barrier contract. Its size is $N = n \cdot 400$ for $n = 1, 2, 3$. The computation time is about half a second for $n = 1$, five seconds for $n = 2$ and seventeen seconds for $n = 3$ on the same hardware as in Table 1. We also ran the algorithm for $N = 1600$ and obtained identical results (up to four decimals) as the ones in the column $N = 1200$.

Representation (7.9) of the jump measure $\nu(x, dy)$ of the asset price process S can now be used to construct the “jump” generator Λ_J defined in equations (5.11) and (5.12). Furthermore it is clear from (7.9) that the instantaneous variance term caused by the jumps of the process S in (5.16) is of the form

$$\int_{-1}^{\infty} y^2 \nu(x, dy) = (x/S_0)^\beta 2\lambda \left(\frac{p}{(\eta_1 - 1)(\eta_1 - 2)} + \frac{1 - p}{(\eta_2 + 1)(\eta_2 + 2)} \right) \quad \text{if } \eta_1 > 2.$$

Note that condition (5.13) that the instantaneous variance is finite is satisfied in the context of the present model if $\eta_1 > 2$. The numerical results of the MG algorithm applied to an up-and-in call option are contained in Table 5, and are compared in the case $\beta = 0$ with the corresponding results of Kou and Wang [40] (see Table 3 in [40]).

Figure 8 presents the pricing error for the up and in call option described in Table 5. We plot the error and the computation time for the pricing algorithm in the model given in (7.6) for $\lambda = 3$ and $\beta = -1$. Unlike in the case of double knock-out options where a truncation error can be avoided, a truncation error will be present when pricing a single up-and-in option, as in the example above. To ensure that the truncation error is sufficiently small we ran the algorithm for an increasing sequence of values of x_N , the largest point in the state-space $\mathbb{G}$ of the chain, and observed that the outcomes did not change up to the required accuracy. The slope is approximately -2 suggesting a quadratic rate of convergence in the spatial mesh size.

7.3. Time-inhomogeneous jump-diffusions.

7.3.1. *Time-inhomogeneous CEV-Kou model.* Let L be a compound Poisson process with intensity λ and jumps of the form $(e^K - 1)$, where K is a normal random variable with mean m and variance

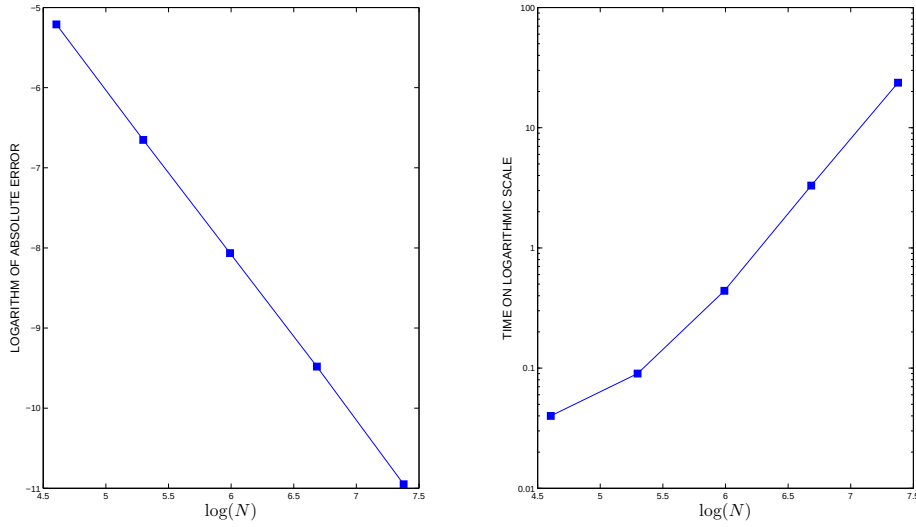


FIGURE 8. The first figure displays the logarithm of the absolute pricing errors for the MG algorithm applied to the model in (7.6) as a function of $\log(N)$, where N is the number of states in the grid. The option under consideration is the up and in call option specified in Table 5. We take the value 9.768837, obtained by running the MG algorithm for $N = 5000$, to be the correct prices in model (7.6) given by the parameter values $\lambda = 3$ and $\beta = -1$. In order to ensure that the truncation error is under control we computed the option value with the largest grid point equal to 300 and 600 (for $N = 5000$) and obtained the same price as above in the first six decimal points. The second figure shows the computation time (in seconds) of the algorithm as a function of $\log(N)$.

s^2 . Consider an asset price process S which starts at $S_0 \in \mathbb{E}$ and is given by the SDE

$$(7.10) \quad \frac{dS_t}{S_{t-}} = (r(t) - \lambda\zeta)dt + \Sigma(t, S_{t-})dW_t + dL_t, \quad \text{where}$$

$$\zeta := \mathbf{E}[e^K - 1] = e^{m+s^2/2} - 1$$

and the Brownian motion W is independent of L . The volatility function $\Sigma(t, x)$ and the instantaneous interest rate $r(t)$ are given by

$$(7.11) \quad \Sigma(t, x) := v(t)(x/S_0)^\beta, \quad \text{where } v(t) := \theta + (\sigma_0 - \theta)e^{-kt},$$

$$(7.12) \quad r(t) := r_0 + r_1 e^{a_0 t}.$$

The constant $v(0) = \sigma_0$ is the starting value of the volatility in our model while the constant θ represents the level to which the function v tends in time. The process S can be viewed as a local volatility Merton jump diffusion. Such processes were used in [3] and [31] for calibration purposes. The parametric form of the time-dependent local volatility function was taken from [48], Section 10.7.

The Lévy measure of L is of the form

$$(7.13) \quad \nu(dy) = \frac{\lambda}{\sqrt{2\pi s^2}} \exp\left(-\frac{(\log(1+y) - m)^2}{2s^2}\right) \frac{dy}{1+y}, \quad y \in (-1, \infty).$$

Time-dependent jump-diffusion		$n = 10$	$n = 100$
$N = 400$	Rebate: $\ell = 90, u = 120$	0.8412	0.8407
	Double-no-touch: $\ell = 90, u = 120$	0.1368	0.1372
$N = 800$	Rebate: $\ell = 90, u = 120$	0.8412	0.8407
	Double-no-touch: $\ell = 90, u = 120$	0.1367	0.1372

TABLE 6. The model is jump-diffusion (7.10) with volatility function (7.11), time-dependent drift (7.12) and normally distributed jumps (7.13). The constants in the volatility and drift function are given by $S_0 = 100$, $\sigma_0 = 0.25$, $\theta = 0.2$, $k = 10$, $\beta = -2$ and $r_0 = 0.01$, $r_1 = 0.09$, $a_0 = -1$. The function $r(t)$ in (7.12) represents the instantaneous time-dependent interest rate (the dividend yield equals $d(t) = 0$). The jump parameters $\lambda = 0.1008$, $m = -0.9144$ and $s = 0.4367$ are taken from the calibrated model in [31]. The state-space of the approximating chain is defined by the algorithm in Section 5 with the same parameters as in Table 2. N is the number of states and n stands for the number of time steps. The table contains the prices of the double-no-touch option that pays one at expiry if the barriers ℓ and u have not been breached, and a rebate option (i.e. American knock-in option) that pays one when the spot leaves the corridor between the barriers. All options expire in $T = 0.5$ years. The algorithm was also run for $n = 500$ time steps and $N = 1600$ states and the results obtained were identical (up to four decimals) to the ones in the $n = 100$ column.

If Φ denotes the standard normal cdf we find that if $-1 \leq a < b$ then

$$\begin{aligned} \int_a^b \nu(dy) &= \lambda [\Phi((\log(1+b) - m)/s) - \Phi((\log(1+a) - m)/s)] \quad \text{and} \\ \int_{-1}^{\infty} y^2 \nu(dy) &= \lambda \left[e^{2(m+s^2)} - 2e^{m+s^2/2} + 1 \right]. \end{aligned}$$

We can therefore apply directly the discretization algorithm described in Section 5.3 to obtain the generator of the approximating chain X during each time interval of constancy. A straightforward application of Corollary 1 yields the results in Table 6.

7.3.2. *VG-Sato process.* Carr et al. [12] introduced into financial modelling the exponential *Sato process*. A Sato process is an additive process $X = \{X_t\}_{t \geq 0}$, which is *self-similar* (i.e. $X_t \sim t^\gamma X_1$ for some constant $\gamma > 0$ and all $t > 0$) and whose law at time one is *self-decomposable*. In Theorem 1 of [12] it is proved that the characteristic function of X_t is of the form

$$(7.14) \quad \Phi_X(u, t) = \mathbf{E}_0 [e^{iuX_t}] = \exp \left(\int_{\mathbb{R}} (e^{iuy} - 1) \frac{h(y/t^\gamma)}{|y|} dy \right)$$

where the function $h : \mathbb{R} \rightarrow \mathbb{R}_+$ satisfies the conditions

$$h(\pm x_1) \geq h(\pm x_2) \quad \text{for all } 0 < x_1 \leq x_2 \quad \text{and} \quad \int_{\mathbb{R}} \min\{x^2, |e^x - 1|\} \frac{h(x)}{|x|} dx < \infty.$$

In particular X is an additive process with representation (4.4) given by $\beta \equiv 0$, $\sigma \equiv 0$ and the density of the compensator given by

$$g(t, x) = \frac{\gamma}{t^{1+\gamma}} \begin{cases} -h'(x/t^\gamma), & x > 0, \\ h'(x/t^\gamma), & x < 0. \end{cases}$$

VG-Sato process	$n = 50$	$n = 500$
N=1600	0.2301	0.2304
N=2000	0.2302	0.2305
N=2400	0.2303	0.2306
N=2800	0.2304	0.2307
N=3200	0.2304	0.2307

TABLE 7. The parameter values for the VG-Sato process are taken from [12] where the model was calibrated to options on the Amazon stock. The values are $\nu = 0.7077, \gamma = 0.4465, \theta = -1.13540, \sigma = 0.7721$ and the parameters C, G, M in (7.15) are given by the formulas $1/C = \nu$, $1/G = (\sqrt{\theta^2 \nu^2/4 + \sigma^2 \nu/2} - \theta \nu/2)$, $1/M = (\sqrt{\theta^2 \nu^2/4 + \sigma^2 \nu/2} + \theta \nu/2)$. The market data and contract details for the double-no-touch option are $S_0 = 100, r = 0.02, d = 0, T = 1/12$ and $\ell = 80, u = 120$. The state-space of the approximating chain contains N points and n denotes the number of time-steps. We also computed the double-no-touch prices in the case $n = 2000$ for all N reported in this table and found that the numbers are the same (the first four decimals) as the ones in the column $n = 500$.

In the specific case of the VG-Sato process we take the function h in (7.14) to equal

$$(7.15) \quad h(x) := C \left(\exp(-G|x|) \mathbf{I}_{(-\infty, 0)} + \exp(-Mx) \mathbf{I}_{(0, \infty)} \right), \quad \text{where } M > 2T^\gamma, C, G > 0$$

and T is the maturity of interest. Note that the analogue of the integral in (5.11) in the current setting can be computed in terms of the function h as:

$$\int_u^v g(t, y) dy = \frac{\gamma}{t} (h(u/t^\gamma) - h(v/t^\gamma)) \quad \text{for } 0 < u \leq v.$$

An analogous expression holds if $u \leq v < 0$. Numerical results are contained in Table 7.

8. CONCLUSION

In this paper we presented an algorithm for pricing barrier options in one-dimensional Markovian models based on an approximation by continuous-time Markov chains. The generator of the approximating chain is constructed by the matching of instantaneous moments of the infinitesimal generator of the Markov process in question, on a suitable non-uniform grid. The approximate barrier option prices are then obtained by calculating the corresponding first-passage distributions for the approximating Markov chain.

To illustrate the flexibility of the method we implemented the algorithm for a number of models, including local volatility models with jumps and models with time-dependent jump-distributions. In the cases of the diffusion and jump-diffusion models where results had been obtained before in the literature, the algorithm produced outcomes that accurately matched those results, and we numerically investigated the order of decay of the error.

We provided a mathematical proof of the convergence of the outcomes of the algorithm to the true prices and derived error estimates under additional regularity assumptions. We derived a theoretical upper bound for the error of the outcomes produced by the algorithm that is linear in the spatial mesh size and the truncation error. We showed that an additional logarithmic

factor may arise in this error bound when the Lévy density has a pole of order two at the origin. In addition, this bound is also linear in the time mesh size if the model is time-inhomogeneous. Numerical experiments suggest that for a number of models the error of the outcomes generated by the algorithm actually decays quadratically in the spatial mesh-size. It would be of interest to establish error bounds under weaker regularity assumptions, and obtain sharp rates of convergence for the specific models, which is a topic left for future research.

Although in principle the method also applies to higher-dimensional Markov processes, the size of the generator matrix would make straightforward application of the algorithm computationally infeasible. The investigation of efficient extensions of the approach to Markov processes of moderate dimension is another topic left for future research.

APPENDIX A. PROOFS

A.1. Proof of Lemma 1. We will restrict to the case $r = 0$, and will leave the extension to the case $r > 0$ to the reader. To prove the convergence in Eqn. (2.12) we need to show that, for $f \in \mathcal{D} \subset C_0(\mathbb{E})$, $g_t(x) \rightarrow 0$ as $t \downarrow 0$, where

$$g_t(x) := t^{-1}(\mathbf{E}_x[f(S_t)\mathbf{I}_{\{t < \tau_A\}}] + \mathbf{E}_x[f(S_{\tau_A})\mathbf{I}_{\{t \geq \tau_A\}}] - f(x)) - \tilde{k}(x),$$

and $\tilde{k}$ is defined in (2.12). By definition, $g_t(x) = 0$ for $x \in A$ whereas for $x \in \mathbb{E} \setminus A$,

$$\begin{aligned} g_t(x) &= t^{-1}(\mathbf{E}_x[f(S_t)\mathbf{I}_{\{t < \tau_A\}}] + \mathbf{E}_x[f(S_{\tau_A})\mathbf{I}_{\{t \geq \tau_A\}}] - f(x)) - \mathcal{L}f(x) \\ (A.1) \quad &= \{t^{-1}(\mathbf{E}_x[f(S_t)] - f(x)) - \mathcal{L}f(x)\} - \{t^{-1}\mathbf{E}_x[(f(S_t) - f(S_{\tau_A}))\mathbf{I}_{\{t \geq \tau_A\}}]\}, \end{aligned}$$

which tends to zero as $t \downarrow 0$. Indeed, note that the first term in (A.1) tends to zero since

$$(A.2) \quad C_t := \sup_{x \in \mathbb{E}} |t^{-1}(\mathbf{E}_x[f(S_t)] - f(x)) - \mathcal{L}f(x)| \rightarrow 0 \quad \text{as } t \downarrow 0.$$

Furthermore, the strong Markov property and the fact that $P_0f = f$ imply that the second term tends to zero, which can be seen as follows. By the triangle inequality,

$$(A.3) \quad |t^{-1}\mathbf{E}_x[\mathbf{I}_{\{t \geq \tau_A\}}(P_{t-\tau_A}f(S_{\tau_A}) - f(S_{\tau_A}))]| \leq \mathbf{P}_x(\tau_A < t) \sup_{s \leq t, y \in \mathbb{E}} s^{-1}|P_s f(y) - f(y)|,$$

which tends to zero as $t \downarrow 0$, since the second term is bounded in view of (A.2).

To establish the convergence in Eqn. (2.15) we note that for $g \in \mathcal{D}$

$$\begin{aligned} &t^{-1}(\mathbf{E}_x[g(S_t)\mathbf{I}_{\{t < \tau_A\}}] - g(x)) - \mathcal{L}g(x) \\ (A.4) \quad &= \{t^{-1}(\mathbf{E}_x[g(S_t)] - g(x)) - \mathcal{L}g(x)\} - \{t^{-1}\mathbf{E}_x[\mathbf{I}_{\{t \geq \tau_A\}}(P_{t-\tau_A}g(S_{\tau_A}))]\}. \end{aligned}$$

As $t \downarrow 0$, the first term in (A.4) tends to zero because of the definition of $\mathcal{L}g$, as in (A.2), while the second term tends to zero if the first condition in Eq. (2.14) holds since g is bounded. If the second condition in Eq. (2.14) holds, the estimate in (A.3) applies with f replaced by g .

Note that if $f \in \mathcal{D}$ is such that $\lim_{x \rightarrow \partial A} \mathcal{L}f(x) = 0$, then $\tilde{k} \in C_0(\mathbb{E})$. For such an f , if $\tilde{S}^A$ is a Feller process, the point-wise convergence for every $x \in \mathbb{E}$ of $t^{-1}(\tilde{P}_t^A f(x) - f(x))$ to $\tilde{k}(x)$ as $t \downarrow 0$ implies that $\tilde{\mathcal{L}}^A f = \tilde{k}$, where $\tilde{\mathcal{L}}^A$ is the generator of $\tilde{P}^A$. This fact follows as a consequence of the Hille-Yosida theorem, see e.g. [57, Lemma 31.7]. A similar argument applies to $\tilde{\mathcal{L}}g$.

A.2. Proof of the Feller property and the form of the generator of an exponential additive process. Let S be an exponential additive process as given in (4.3)–(4.4) and let (D, Y) be as in (4.1). In view of the right continuity of $t \mapsto (D_t, S_t)$, the boundedness of the function $(s, x) \mapsto f(s, x)$ and the dominated convergence theorem, it follows that $Q_t f(s, x)$ converges to $f(s, x)$ for every $s \in [0, T]$ and $x \in \mathbb{E}$ as $t \downarrow 0$. The linear function $g \mapsto Q_t g$ maps $C_0([0, T] \times \mathbb{E})$ to itself, which follows by combining the dominated convergence theorem, the spatial homogeneity of X and the fact that $g \in C_0([0, T] \times \mathbb{E})$. Thus, (D, S) is a Feller process.

Denote the marginal distribution and corresponding characteristic function of the increment $X_{t+s} - X_s$ by $\mu_{s,s+t}$ and $\hat{\mu}_{s,s+t}(z)$ respectively. In view of the form (4.4) and the independent increments property of X it holds that

$$\begin{aligned}\hat{\mu}_{s,s+t}(z) &= \int_s^{s+t} \psi_u(z) du \quad \text{where} \\ \psi_u(z) &:= iz\beta(u) - \frac{z^2}{2}\sigma(u)^2 + \int_{\mathbb{R} \setminus \{0\}} [e^{izy} - 1 - izy]g(u, y)dy.\end{aligned}$$

The characteristic function $\hat{\mu}_n$ of a compound Poisson process with Lévy measure $t_n^{-1}\mu_{s,s+t_n}$ is given by

$$\begin{aligned}\hat{\mu}_n(z) &= \exp\left(\frac{1}{t_n} \int_{\mathbb{R}} (e^{izx} - 1) \mu_{s,s+t_n}(dx)\right) \\ &= \exp(t_n^{-1}(\hat{\mu}_{s,s+t_n}(z) - 1)) \\ &= \exp\left(t_n^{-1}(e^{\int_s^{s+t_n} \psi_u(z) du} - 1)\right).\end{aligned}$$

For fixed z and any sequence $(t_n)_{n \in \mathbb{N}}$ that converges monotonically to zero, we find that, as $n \rightarrow \infty$,

$$\hat{\mu}_n(z) \rightarrow \exp(\psi_s(z)) =: \hat{\mu}_s(z),$$

since $t \mapsto \int_0^t \psi_{s+u}(z) du$ is right differentiable at zero with derivative $\psi_s(z)$, in view of the dominated convergence, the integrability condition (4.5) and the continuity of β, σ and $g(\cdot, y)$. Here $\hat{\mu}_s(z)$ denotes a characteristic function of an infinitely divisible distribution μ_s . Hence an argument analogous to that in the proof of Theorem 31.5 in Sato [57] can be constructed to verify that for $f \in C_c^2(\mathbb{R}_+)$ the following identity holds

$$\begin{aligned}\lim_{t \downarrow 0} t^{-1}(Q_t f - f)(s, x) &= \tilde{\beta}(s)x f'(x) + \frac{\sigma^2(s)x^2}{2} f''(x) \\ &\quad + \int_{\mathbb{R}} [f(xe^z) - f(x) - \mathbf{1}_{\{|z| < 1\}} f'(x)(e^z - 1)]g(s, z)dz,\end{aligned}$$

where the limit is taken in the strong sense.

A.3. Proof of weak convergence. In this proof we shall employ standard convergence results for Markov processes that can be found in Ethier & Kurtz [23].

The proof is based on the following results. Let $P_t f(x) = \mathbf{E}_x[f(X_t)]$ and $P_t^{(n)} f_n(x) = \mathbf{E}_x[f_n(X_t^{(n)})]$ where, for any function $f : \mathbb{E} \rightarrow \mathbb{R}$, we write f_n to denote $f_n = f|_{\mathbb{G}^{(n)}}$. Denote by $\tilde{\mathcal{D}}(\mathbb{E})$ a core of

the infinitesimal generator $\mathcal{L}$, and write $g_n \rightarrow g$ if $\sup_{x \in \mathbb{G}(n)} |g_n(x) - g(x)| \rightarrow 0$ as $n \rightarrow \infty$. Then it holds that (Ethier and Kurtz [23, Theorem 1.6.1])

$$(A.5) \quad \Lambda^{(n)} f_n \rightarrow \mathcal{L}f \quad \text{for all } f \in \tilde{\mathcal{D}}(\mathbb{E})$$

implies that

$$(A.6) \quad P_t^{(n)} f_n \rightarrow P_t f \quad \text{for all } f \in C_0(\mathbb{E}) \text{ and } t \geq 0.$$

If (6.4)(i) holds, Remark 2 implies that (A.5) (and hence (A.6)) holds. If (6.4)(ii) holds, then [23, Corollary 4.8.7] implies that (A.6) holds.

If (A.6) holds, then for any starting point $S_0 = X_0^{(n)} = x \in \mathbb{E}$, $n \in \mathbb{N}$, we have $X^{(n)} \Rightarrow S$, that is, $X^{(n)}$ converges weakly to S on $D(\mathbb{R})$, the Skorokhod space of càdlàg real-valued functions endowed with the Skorokhod topology (Ethier and Kurtz [23, Theorem 4.2.11]).

Thus, condition (6.3) implies that $X^{(n)} \Rightarrow S$, or, equivalently, for any continuous bounded function $g : D(\mathbb{R}) \rightarrow \mathbb{R}$ it holds that

$$(A.7) \quad \mathbf{E}_x[g(X^{(n)})] \rightarrow \mathbf{E}_x[g(S)] \quad \text{as } n \rightarrow \infty.$$

In fact (A.7) holds if $g : D(\mathbb{R}) \rightarrow \mathbb{R}$ is bounded and continuous on some subset C of $D(\mathbb{R})$ that satisfies $\mathbf{P}_x[S \in C] = 1$ (see Jacod and Shiryaev [36, Section VI.3a]).

To complete the proof we thus have to establish the continuity of the barrier payoff, which is for any $\omega \in D(\mathbb{R})$ given by

$$\omega \mapsto g(\omega) = e^{-r(\tau_A(\omega) \wedge T)} f(\omega(\tau_A(\omega) \wedge T)),$$

in the Skorokhod topology. We refer to Jacod and Shiryaev [36] for background on the Skorokhod topology. Define

$$\begin{aligned} T_{\ell,u}(\omega) &:= \inf\{t \geq 0 : \omega(t) \text{ or } \omega(t-) \notin (\ell, u)\}, \\ T_{\ell,u}^+(\omega) &:= \inf\{t \geq 0 : \omega(t) \notin [\ell, u]\}, \\ J(\omega) &:= \{s \in \mathbb{R}_+ : \omega(s) \neq \omega(s-)\}, \\ V(\omega) &:= \{y = (y_1, y_2) : T_y(\omega) < T_y^+(\omega)\}, \\ V'(\omega) &:= \{y = (y_1, y_2) : T_y(\omega) \in J(\omega), \omega(T_y(\omega)-) \in \{y_1, y_2\}\}. \end{aligned}$$

Then the following general result holds true:

Lemma 2. (a) At every ω such that $t \notin J(\omega)$ the map $D(\mathbb{R}) \rightarrow \mathbb{R}^2$ given by

$$\omega \mapsto \left(\inf_{0 \leq s \leq t} \omega(s) \wedge 0, \sup_{0 \leq s \leq t} \omega(s) \vee 0 \right)$$

is continuous.

(b) At every ω such that $(\ell, u) \notin V(\omega)$ and $T_{\ell,u}(\omega) < \infty$ the map $D(\mathbb{R}) \rightarrow \mathbb{R}_+$ given by

$$\omega \mapsto T_{\ell,u}(\omega)$$

is continuous.

(c) At every ω such that $t \notin J(\omega)$ and $(\ell, u) \notin V(\omega) \cup V'(\omega)$ the map $D(\mathbb{R}) \rightarrow \mathbb{R}$ given by

$$\omega \mapsto \omega(T_{\ell,u}(\omega) \wedge t)$$

is continuous.

The proofs of Lemma 2(a), (b) and (c) are straightforward adaptations of Propositions VI.2.4, VI.2.10 and VI.2.11 in Jacod and Shiryaev [36].

Assume now that the process S is a coordinate process on the canonical probability space $D(\mathbb{R})$, i.e. $\omega = S(\omega)$ for each $\omega \in D(\mathbb{R})$. The fact that S is quasi-left continuous (as it is a Feller process, e.g. [13, Theorem 2.4]) implies that at each fixed time t the path $t \mapsto S_t(\omega)$ is continuous almost surely, that is, $\mathbf{P}_x(t \notin J(\omega)) = 1$ for any $x \in \mathbb{E}$. Furthermore, quasi-left continuity also implies that $\mathbf{P}_x[(y_1, y_2) \in V'(\omega)] = \mathbf{P}_x[T_y(\omega) \in J(\omega), \omega(T_y(\omega)-) \in \{y_1, y_2\}] = 0$ for any pair y_1, y_2 . Indeed, on the event $\{\omega(T_y(\omega)-) \in \{y_1, y_2\}, T_y(\omega) < \infty\}$, it holds that $\omega(T_y(\omega)-) = \omega(T_y(\omega)) \in \{y_1, y_2\}$ almost surely, as for any increasing sequence of stopping times $T_n < T_y$ converging to T_y it holds that $\omega(T_n) \rightarrow \omega(T_y-) = \omega(T_y)$ almost surely on $\{T_y(\omega) < \infty\}$. Thus, almost surely $T_{\ell,u}$ is equal to

$$\tilde{T}_{\ell,u}(\omega) := \inf\{t \geq 0 : \omega(t) \notin (\ell, u)\}.$$

Further, the condition (6.2) implies that

$$\mathbf{P}_x(\tau_A = T) = 0.$$

Indeed, the event $\{\tau_A = T\}$ is equal to the union of the events $\{S_T = S_{\tau_A} \in \{\ell, u\}\}$, and $\{S_T = S_{\tau_A} \in (0, \ell) \cup (u, \infty)\}$, both of which have zero probability; the former by condition (6.2) and the latter since $\mathbf{P}_x(S_T \neq S_{T-}) = 0$ by quasi-left continuity of S .

Moreover, note that Assumption 2 implies that $\mathbf{P}_x((\ell, u) \notin V(\omega)) = 0$. The proof of Theorem 2 is then completed by combining the foregoing with the convergence in equation (A.7) and Lemma 2.

A.4. Proof of time-dependent convergence. Theorem 2 remains valid for a higher-dimensional Feller process if the first passage is defined on one of the coordinates, as the above proof carries over. The question we need to answer is how to construct the sequence of Markov chains that approximates the two-dimensional Feller process (D, Y) (see (4.1) for definition) in such a way that the limits in Corollary 2 hold.

A.4.1. Construction of an approximating two-dimensional time-homogeneous chain. To fit the case of a time-inhomogeneous Markov process into this framework we will consider an approximation of the time-space process (D, Y) . We approximate the time-space Feller Markov process (D, Y) by a two-dimensional (time-homogeneous) Markov chain (Z, X) living on a time-space grid $\mathbb{T} \times \mathbb{G}$ where $\mathbb{T} = \{0\} \cup \delta_m \mathbb{N}$ with $\delta_m = T/(nm)$ and $n, m \in \mathbb{N}$. To define this chain, let $\mathcal{L}'$ be the infinitesimal generator of (D, Y) and $\mathcal{L}_t f$ the corresponding space restricted generator. Assume that $\Lambda^{(i)}$, $i = 1, \dots, n$ are generator matrices approximating the generator $\mathcal{L}_{iT/n}$. Then the generator of the two-dimensional chain (Z, X) is specified as

$$(A.8) \quad \Lambda^{(n,m)} f(t, x) = \delta_m^{-1} (f(t + \delta_m, x) - f(t, x)) + \sum_{i=1}^n (\Lambda^{(i)} f_t)(x) \mathbf{I}_{\{(i-1)m\delta_m \leq t < im\delta_m\}},$$

for $(t, x) \in \mathbb{T} \times \mathbb{G}$, where $f_t(x) = f(t, x)$, for any function $f : \mathbb{T} \times \mathbb{G} \rightarrow \mathbb{R}$.

The corresponding stochastic dynamics are described as follows. The first component is given by $Z_t = \delta_m \cdot t$, δ_m times a standard Poisson process N with rate $\lambda = 1/\delta_m$. Thus, Z is piecewise

constant and moves by positive jumps of size δ_m that occur after independent exponential times with mean λ . For any $t > 0$, Z_t follows a Poisson distribution with mean t and variance $\delta_m t = tT/(nm)$. Hence, as $n \rightarrow \infty$, Z_t tends to t for every fixed $t > 0$. In fact, since $\{Z_t - t\}_{t \geq 0}$ is a martingale, Doob's maximal inequality implies that

$$\mathbf{E} \left[\sup_{t \leq T} (Z_t - t)^2 \right] \leq \text{Var}[Z_T] = T^2/(nm),$$

so that $\{Z_t\}_{t \in [0, T]}$ converges uniformly to the deterministic unit drift $\{D_t\}_{t \in [0, T]}$, almost surely. Conditional on Z_t taking a value in $[(i-1)\delta_m, i\delta_m)$, X evolves as a Markov chain with state-space $\mathbb{G}$ and generator matrix $\Lambda^{(i)}$.

In this setting we consider the barrier option with a *randomised* time of maturity

$$T^{(nm)} = \inf\{t \geq 0 : Z_t = T\},$$

the first time that Z hits the level $T = mn\delta_m$, which is equal to the nm -th jump time of Z . Thus $T^{(nm)}$ is distributed as the sum of mn independent exponential random variables with mean $T/(nm)$, and follows a Gamma distribution with mean T and variance $T^2/(nm)$. The corresponding (stochastic) discounting is defined as

$$(A.9) \quad R_t = \int_0^t \rho(Z_s) ds, \quad \text{where} \quad \rho(u) = \sum_{i=1}^n r_i \mathbf{I}_{\{(i-1)m\delta_m \leq u < im\delta_m\}}.$$

The idea of randomising the maturity was employed before by Carr [9] to approximate a finite maturity American put option by replacing its maturity by an independent $\text{Gamma}(n, T/n)$ random variable, and this technique has also been employed to value barrier-type options (see e.g. [6]).

Corollary 3. *For any $T > 0$, ρ in (A.9) and any functions $\phi : \widehat{\mathbb{G}} \rightarrow \mathbb{R}$ and $\psi : \mathbb{G} \rightarrow \mathbb{R}$ with $\psi(x) = 0$ for $x \in \widehat{\mathbb{G}}$, it holds that*

$$(A.10) \mathbf{E}_x \left[\phi(X_{T^{(nm)}}) \mathbf{I}_{\{\tau > T^{(nm)}\}} \right] = \left[\left(I - \delta_m \widehat{\Lambda}^{(1)} \right)^{-m} \cdots \left(I - \delta_m \widehat{\Lambda}^{(n)} \right)^{-m} \phi \right] (x), \quad x \in \widehat{\mathbb{G}},$$

$$(A.11) \mathbf{E}_x \left[e^{-R_\tau} \psi(X_\tau) \mathbf{I}_{\{\tau < T^{(nm)}\}} \right] = \left[\left(I - \delta_m \widetilde{\Lambda}_{r_1}^{(1)} \right)^{-m} \cdots \left(I - \delta_m \widetilde{\Lambda}_{r_n}^{(n)} \right)^{-m} \psi \right] (x), \quad x \in \mathbb{G},$$

where $\delta_m = T/(nm)$ and the stopping time τ is defined in (3.5).

Proof. The strong Markov property of (Z, X) at the stopping time $T^{(1)}$ yields that

$$\mathbf{E}_x \left[\phi(X_{T^{(nm)}}) \mathbf{I}_{\{\tau > T^{(nm)}\}} \right] = \mathbf{E}_x \left[\mathbf{I}_{\{\tau > T^{(1)}\}} \mathbf{E}_{X_{T^{(1)}}} \left[\mathbf{I}_{\{\tau > T^{(mn-1)}\}} \phi(X_{T^{(mn-1)}}) \right] \right],$$

since $T^{(i)}$ is in distribution equal to the sum of i independent exponential random variables. It follows from Theorem 1 that for any $f : \mathbb{G} \rightarrow \mathbb{R}$

$$\begin{aligned} \mathbf{E}_x [\mathbf{I}_{\{\tau > T^{(1)}\}} f(X_{T^{(1)}})] &= \int_0^\infty \lambda e^{-\lambda t} \mathbf{E}_x [\mathbf{I}_{\{\tau > t\}} f(X_t)] dt \\ &= [(I - \lambda^{-1} \widehat{\Lambda}^{(1)})^{-1} f](x), \quad \text{where } \lambda = \delta_m^{-1}. \end{aligned}$$

The expression (A.10) then follows by induction. The proof of the identity (A.11) is similar and is therefore omitted. $\square$

A.4.2. Proof of Corollary 2. For any sequence $(m(n))_{n \in \mathbb{N}}$ of natural numbers, let $(Z^{(m,n)}, X^{(n)})_{n \in \mathbb{N}}$ be a sequence of continuous-time Markov chains with state-spaces $\mathbb{T}^{(n)} \times \mathbb{G}^{(n)}$ and generator matrices $\Lambda^{(n,m)}$ as constructed in the previous section, such that their generators uniformly converge to the two-dimensional generator $\mathcal{L}'$ of (D, Y) , in the sense that

$$(A.12) \quad \epsilon_{n,m}(f) = \max_{t \in \mathbb{T}^{(n)}, x \in \mathbb{G}^{(n)}} \left| \Lambda^{(n,m)} f_n(t, x) - \mathcal{L}' f(t, x) \right| \rightarrow 0$$

for all functions f in a core of $\mathcal{L}'$, where $f_n = f|_{[0,T] \times \mathbb{G}^{(n)}}$. In view of the form of the generator (A.8) and the condition (6.7) in Corollary 2, it follows that (A.12) holds true. If S satisfies (6.2) and Assumption 2 (with $\mathbf{P}_x$ replaced by $\mathbf{P}_{0,x}$), then the conclusions of Theorem 2 also apply to the two-dimensional Feller process (D, Y) and the sequence $(Z^{(m,n)}, X^{(n)})_{n \in \mathbb{N}}$ (since the proof remains valid for this case). As a consequence the expectations (A.10)–(A.11) converge to the barrier option prices under the limiting model (D, Y) . To complete the proof we must check that the sequences (A.10)–(A.11) converge to the same limits as (4.10)–(4.12).

The latter follows by observing that, for a given $m \in \mathbb{N}$, a matrix Λ , a maturity $T > 0$ and $\delta_m = T/(nm)$ it holds that,

$$\begin{aligned} \exp \left\{ \frac{T}{n} \Lambda \right\} &= I + \frac{T}{n} \Lambda + \frac{T^2}{2n^2} \Lambda^2 + o(n^{-3}) \\ (I - \delta_m \Lambda)^{-m} &= I + \frac{T}{n} \Lambda + \frac{T^2}{2n^2} \Lambda^2 + \frac{T^2}{2n^2 m} \Lambda^2 + o(n^{-3}) \end{aligned}$$

as n tends to infinity. Hence, for given matrices $\Lambda^{(1)}, \dots, \Lambda^{(n)}$, we have that

$$(A.13) \quad \left\| \exp \left\{ \frac{T}{n} \Lambda^{(1)} \right\} \cdots \exp \left\{ \frac{T}{n} \Lambda^{(n)} \right\} - (I - \delta_m \Lambda^{(1)})^{-m} \cdots (I - \delta_m \Lambda^{(n)})^{-m} \right\| \leq \frac{C^{(n)}}{n^2 m}$$

where $C^{(n)}$ is some constant that depends on $\Lambda^{(1)}, \dots, \Lambda^{(n)}$ and $\|\cdot\|$ is a matrix norm.

Take the sequence of natural numbers $(m(n))_{n \in \mathbb{N}}$ such that it satisfies $m(n) \geq C^{(n)}$ for all $n \in \mathbb{N}$, where $(C^{(n)})_{n \in \mathbb{N}}$ are the constants in (A.13), that correspond to the sequence $(Z^{(m,n)}, X^{(n)})$. It then follows from (A.13) that the expectations (A.10)–(A.11) and (4.10)–(4.12) converge to the same limit as $n \rightarrow \infty$.

A.5. Explicit construction of the generator matrices. The action of $\Lambda_D^{(j)}$, which discretizes the operator (6.10), on a function $f : \mathbb{G}^{(n)} \rightarrow \mathbb{R}$ is described by

$$(A.14) \quad \Lambda_D^{(j)} f(x_i) = \begin{cases} \frac{\sigma_{ij}^2}{2} \Delta f(x_i) + (\gamma_{ij})_+ \nabla^+ f(x_i) + (\gamma_{ij})_- \nabla^- f(x_i) & i \in \{2, \dots, n-1\}, \\ 0 & i \in \{1, n\}, \end{cases}$$

where $y_{\pm} = \max\{\pm y, 0\}$ and $\mathbb{G}^{(n)}$ is defined in (6.16). We write x_i instead of $x_i^{(n)}$ and will do so in all that follows. We denote by $\nabla^{\pm}$ and Δ the first and second order difference operators given by

$$(A.15) \quad \nabla^- f(x_i) = \frac{1}{x_i - x_{i-1}} [f(x_i) - f(x_{i-1})],$$

$$(A.16) \quad \nabla^+ f(x_i) = \frac{1}{x_{i+1} - x_i} [f(x_{i+1}) - f(x_i)],$$

$$(A.17) \quad \Delta f(x_i) = \frac{2}{x_{i+1} - x_{i-1}} [\nabla^+ f - \nabla^- f](x_i),$$

with σ_{ij}^2 and γ_{ij} defined by

$$\sigma_{ij}^2 := \sigma^2(t_j, x_i)x_i^2, \quad \gamma_{ij} := \gamma(t_j)x_i,$$

where $\sigma^2(t, x)$ and $\gamma(t)$ are as in (6.9). Note that the discretization in (A.14) defines a birth-death process also in the case that $\sigma_{ij} = 0$.

The jump part of the generator, given in (6.11), can be rewritten as follows

$$(A.18) \quad \mathcal{L}_{J,t}f(t, x) = \int_1^\infty [f(t, x(1+y)) - f(t, x) - f'(t, x)xy]g(t, x, y)dy \\ + \begin{cases} \int_{-1}^1 \frac{1}{|y|} [f(t, x(1+y)) - f(t, x)]|y|g(t, x, y)dy \\ \quad - f'(t, x)x \int_{-1}^1 yg(t, x, y)dy, & \mathcal{I}(t, x) < \infty, \\ \int_{-1}^1 \frac{1}{y^2} [f(t, x(1+y)) - f(t, x) - f'(t, x)xy](y^2g(t, x, y))dy, & \mathcal{I}(t, x) = \infty. \end{cases}$$

Note that $\mathcal{I}(t, x)$, defined in (6.13), is either finite for all x and t or infinite for all x and t because we are assuming that the process S we are approximating falls into one of the categories O, I, II. The representation (A.18) has a natural discretization, which we will now describe, because the second integral in (A.18) is an integral of a continuous function against a finite measure.

For different levels of activity of the purely discontinuous part of S , quantified by whether or not $\mathcal{I}_{i,j} = \mathcal{I}(t_j, x_i)$ defined in (6.13) is finite, the matrix $\Lambda_J^{(j)}$ is defined in the following way. Let

$$A_i^0(h) = (-1 + x_{\underline{b}_i}/x_i, -1 + x_{\bar{b}_i}/x_i)$$

where the levels $\underline{b}_i$ and $\bar{b}_i$ are given by

$$(A.19) \quad \bar{b}_i = \min\{j > i : x_j \geq x_i + C_+(h)h\}, \quad \underline{b}_i = \max\{j < i : x_j \leq x_i - C_-(h)h\},$$

with $h = h(n)$ the spatial mesh size and

$$(A.20) \quad C_\pm(h) = d_\pm \cdot c(\alpha_\pm, h), \quad \text{for some constants } d_\pm \text{ satisfying } d_\pm > \frac{\bar{\kappa}_\pm}{\underline{\kappa}_\pm},$$

where $\alpha_\pm$, $\bar{\kappa}_\pm$ and $\underline{\kappa}_\pm$ are constants given in (6.14) and (6.15), and

$$(A.21) \quad c(\alpha, h) = \begin{cases} \frac{2-\alpha}{\alpha-1}, & \alpha \in (1, 2), \\ -\log(h), & \alpha = 1. \end{cases}$$

We can then define

$$c_{ik,j}^{(p)} = \begin{cases} \int_{\mathbb{R}} |y|^p g(t_j, x_i, y) \mathbf{I}_{\left[\frac{x_{k-1}}{x_i}-1, \frac{x_k}{x_i}-1\right]}(y) dy, & k > i, \\ \int_{\mathbb{R}} |y|^p g(t_j, x_i, y) \mathbf{I}_{A_i^0(h)}(y) dy, & k = i, \\ \int_{\mathbb{R}} |y|^p g(t_j, x_i, y) \mathbf{I}_{\left[\frac{x_k}{x_i}-1, \frac{x_{k+1}}{x_i}-1\right]}(y) dy, & k < i. \end{cases}$$

Finally we define the discretized version of the generator as

$$(A.22) \quad \Lambda_J^{(j)} f(x_i) = \sum_{k: x_k > 2x_i} \left[c_{ik,j}^{(0)} (f(x_k) - f(x_i)) - x_i \nabla^- f(x_i) c_{ik,j}^{(1)} \right] + \bar{\Lambda}_J^{(j)} f(x_i), \quad i \notin \{1, n\}, \\ \Lambda_J^{(j)} f(x_i) = 0, \quad i \in \{1, n\}.$$

If $\mathcal{I}_{i,j}$ is finite $\bar{\Lambda}_J^{(j)} f(x_i)$ is given by

$$(A.23) \quad \bar{\Lambda}_J^{(j)} f(x_i) := \sum_{k \neq i: x_k \leq 2x_i} \frac{x_i}{|x_k - x_i|} [f(x_k) - f(x_i)] c_{ik,j}^{(1)} - x_i \alpha_{i,j} \tilde{\nabla} f(x_i)$$

where

$$(A.24) \quad \tilde{\nabla} f(x_i) := \begin{cases} \nabla^- f(x_i) & \alpha_{i,j} \geq 0 \\ \nabla^+ f(x_i) & \alpha_{i,j} < 0 \end{cases} \quad \text{where} \quad \alpha_{i,j} := \int_{-1}^1 yg(t_j, x_i, y) dy.$$

If $\mathcal{I}_{i,j}$ is equal to infinity we have

$$(A.25) \quad \begin{aligned} \bar{\Lambda}_J^{(j)} f(x_i) &:= \frac{1}{2} c_{ii,j}^{(2)} x_i^2 \Delta f(x_i) \\ &+ \sum_{k: (x_k - x_i)/x_i \in (-1, 1) \setminus A_i^0(h)} \left(\frac{x_i}{x_k - x_i} \right)^2 [f(x_k) - f(x_i) - (x_k - x_i) \nabla^0 f(x_i)] c_{ik,j}^{(2)}. \end{aligned}$$

Here $\nabla^0 f$ is the (central) difference operator given by

$$(A.26) \quad \nabla^0 f(x_i) = \frac{x_i - x_{i-1}}{x_{i+1} - x_{i-1}} \nabla^+ f(x_i) + \frac{x_{i+1} - x_i}{x_{i+1} - x_{i-1}} \nabla^- f(x_i).$$

Clearly, the first part in the expression in (A.22) defines the generator matrix of a Markov chain taking values in $\mathbb{G}^{(n)}$. Similarly the expression in (A.23) yields a generator of a Markov chain. In (A.25) the levels $\bar{b}_i$ and b_i are chosen so as to ensure that $\Lambda_J^{(j)}$ defined in (A.22) is in fact a generator.

To verify that (A.25) defines a Markov jump process we need to show that in the matrix $\Lambda_J^{(j)}$ defined in (A.22) all off-diagonal elements are positive. It follows from the definitions of Δf and $\nabla^0 f$ that this is the case if the following condition holds for all $x_i \in \widehat{\mathbb{G}}^{(n)}$:

$$(A.27) \quad \begin{aligned} x_i c_{ii,j}^{(2)} &\geq \max\{(x_i - x_{i-1}) \nu_{i,j}, (x_i - x_{i+1}) \nu_{i,j}\}, \quad \text{where} \\ \nu_{i,j} &:= \sum_{k: (x_k - x_i)/x_i \in (-1, 1) \setminus A_i^0(h)} \frac{x_i}{x_k - x_i} c_{ik,j}^{(2)}, \end{aligned}$$

for all $j = 1, \dots, m$. For h sufficiently small it is easily checked from the definition of $\nu_{i,j}$ that

$$\nu_{i,j} \leq \int_{(-1, 1) \setminus A_i^0(h)} |y| g(t_j, x_i, y) dy,$$

which implies that (A.27) is satisfied if, for any $(t, x) \in (0, T] \times [\ell, u]$,

$$x \int_{A_i^0(h)} y^2 g(t, x, y) dy \geq h \int_{(-1, 1) \setminus A_i^0(h)} |y| g(t, x, y) dy.$$

This condition is satisfied if, for any $(t, x) \in (0, T] \times [\ell, u]$, the following two conditions hold for h sufficiently small

$$(A.28) \quad x \int_{\left(0, \frac{C_+(h)h}{x}\right)} y^2 g(t, x, y) dy \geq h \int_{\left(\frac{C_+(h)h}{x}, 1\right)} yg(t, x, y) dy,$$

$$(A.29) \quad x \int_{\left(-\frac{C_-(h)h}{x}, 0\right)} y^2 g(t, x, y) dy \geq h \int_{\left(-1, -\frac{C_-(h)h}{x}\right)} |y| g(t, x, y) dy.$$

In view of (6.14) it follows that (A.28) holds if the following inequality is satisfied

$$\begin{aligned} \underline{\kappa}_+ \int_{(0, C_+(h)h/x)} y^{1-\alpha_+} dy &\geq h\bar{\kappa}_+ \int_{(C_+(h)h/x, 1)} y^{-\alpha_+} dy \\ \Leftrightarrow \begin{cases} \frac{\underline{\kappa}_+ x}{2-\alpha_+} \left(\frac{C_+(h)h}{x} \right)^{2-\alpha_+} \geq \frac{\bar{\kappa}_+ h}{\alpha_+ - 1} \left(\left(\frac{C_+(h)h}{x} \right)^{1-\alpha_+} - 1 \right) & \text{in case II,} \\ \underline{\kappa}_+ x \left(\frac{C_+(h)h}{x} \right) \geq -\bar{\kappa}_+ h (\log(C_+(h)) + \log(h) - \log(x)) & \text{in case I.} \end{cases} \end{aligned}$$

The latter holds for all h sufficiently small if

$$\begin{cases} C_+(h) \cdot \frac{\underline{\kappa}_+}{2-\alpha_+} \geq \frac{\bar{\kappa}_+}{\alpha_+ - 1}, & \text{in case II,} \\ \liminf_{h \rightarrow \infty} C_+(h)/|\log h| > \bar{\kappa}_+/\underline{\kappa}_+, & \text{in case I.} \end{cases}$$

Clearly, $C_+(h)$, defined in (A.20), satisfies this condition. The fact that Condition (A.29) is satisfied follows by a similar line of reasoning.

A.6. Proof of error estimates. The following lemma is an important auxiliary result for the development of error-estimates. It formalises the intuition that two semi-groups should be “close” if the corresponding infinitesimal generators are “close”.

Suppose that S is a Feller process on the state-space $\mathbb{E}$ with associated semigroup P and corresponding infinitesimal generator $\mathcal{L}$, and let $(X^{(n)})_{n \in \mathbb{N}}$ be a sequence of Markov chains with state-spaces $\mathbb{G}^{(n)}$, semigroups $P^{(n)}$ and corresponding generator matrices $\Lambda^{(n)}$. For any $f \in \mathcal{D}(\mathcal{L})$, the domain of $\mathcal{L}$, consider the following error measure

$$(A.30) \quad \epsilon_n(f) = \left\| \Lambda^{(n)} f - \mathcal{L} f \right\|_n, \quad s \geq 0,$$

where we write $\Lambda^{(n)} f = \Lambda^{(n)} f_n$ and $P^{(n)} f = P^{(n)} f_n$ with $f_n = f|_{\mathbb{G}^{(n)}}$ and

$$(A.31) \quad \|g\|_n = \sup_{x \in \mathbb{G}^{(n)}} |g(x)|$$

for any $g : \mathbb{E} \rightarrow \mathbb{R}$. Then the following estimate holds true:

Lemma 3. *Let $f \in \mathcal{D}(\mathcal{L})$ and $T > 0$, and suppose that there exists a function $h : \mathbb{N} \rightarrow \mathbb{R}_+$ and a function $c_f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that for all $n \in \mathbb{N}$ and $s \geq 0$*

$$\epsilon_n(P_s f) \leq c_f(s) \cdot h(n).$$

Then it holds that

$$\sup_{t \in [0, T]} \left\| P_t f - P_t^{(n)} f \right\|_n \leq \int_0^T c_f(s) ds \cdot h(n).$$

Proof. The proof closely follows that of Lemma 6.2 in Ethier and Kurtz [23]. Note first that

$$\begin{aligned} \frac{\partial}{\partial s} \left(P_{t-s}^{(n)} P_s f \right) &= \Lambda^{(n)} P_{t-s}^{(n)} P_s f - P_{t-s}^{(n)} \mathcal{L} P_s f \\ (A.32) \quad &= P_{t-s}^{(n)} \left(\Lambda^{(n)} P_s f - \mathcal{L} P_s f \right), \end{aligned}$$

as $P_{t-s}^{(n)}$ and $\Lambda^{(n)}$ commute. Since the function in (A.32) is continuous in s , the fundamental theorem of calculus and the fact that $P_0^{(n)}f = f$ imply that

$$P_t f - P_t^{(n)} f = \int_0^t P_{t-s}^{(n)} \left(\Lambda^{(n)} P_s f - \mathcal{L} P_s f \right) ds.$$

Thus, the triangle inequality yields that

$$\begin{aligned} \|P_t f - P_t^{(n)} f\|_n &\leq \int_0^t \|P_{t-s}^{(n)} (\Lambda^{(n)} P_s f - \mathcal{L} P_s f)\|_n ds \\ &\leq \int_0^t \|\Lambda^{(n)} P_s f - \mathcal{L} P_s f\|_n ds \\ &\leq \int_0^T c_f(s) ds \cdot h(n) \end{aligned}$$

for any $t \in [0, T]$, as $c_f(s)$ is non-negative. $\square$

A similar estimate can be derived in the time-inhomogeneous setting. To that end, suppose that (D, Y) is a time-space Feller process with state-space $\mathbb{E}' = [0, T] \times \mathbb{E}$, defined as in Section 4. Denote its semi-group by Q and let $\mathcal{L}'$ be the corresponding infinitesimal generator. The process (D, Y) is approximated by a sequence of time-inhomogeneous Markov chains with infinitesimal generators

$$\Lambda^{(n,m)} := \frac{\partial}{\partial t} + \Lambda_t^{(n,m)}$$

where $\Lambda_t^{(n,m)}$ is given in (6.19). For any $f \in \mathcal{D}(\mathcal{L}')$, consider the following error measure, which is a generalisation of ϵ_n to the time-inhomogeneous setting:

$$\epsilon'_n(f) = \left\| \Lambda^{(n,m)} f - \mathcal{L}' f \right\|_n, \quad s \geq 0,$$

where $\|g\|_n = \sup_{t \in [0, T], x \in \mathbb{G}^{(n)}} |g(t, x)|$ for any $g : [0, T] \times \mathbb{E} \rightarrow \mathbb{R}$, and we write $\Lambda^{(n,m)}(f)(t, x) = \frac{\partial f}{\partial t}(t, x) + \Lambda_t^{(n,m)}(f_n)(t, x)$ with $f_n = f|_{[0, T] \times \mathbb{G}^{(n)}}$, the restriction of f to $[0, T] \times \mathbb{G}^{(n)}$.

Then we have the following generalisation of Lemma 3 to the time-dependent setting:

Lemma 4. *Let $f \in \mathcal{D}(\mathcal{L}')$ and $T > 0$, and suppose that there exists functions $h, \delta, k : \mathbb{N} \rightarrow \mathbb{R}_+$ and functions $c_f, \tilde{c}_f, \bar{c}_f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that for all $n \in \mathbb{N}$ and $s \geq 0$*

$$\epsilon'_n(Q_s f) \leq c_f(s) \cdot h(n) + \bar{c}_f(s) \cdot k(n) + \tilde{c}_f(s) \cdot \delta(m).$$

Then it holds that

$$\sup_{t \in [0, T]} \|Q_t f - Q_t^{(n)} f\|_n \leq \int_0^T c_f(s) ds \cdot h(n) + \int_0^T \bar{c}_f(s) ds \cdot k(n) + \int_0^T \tilde{c}_f(s) ds \cdot \delta(m).$$

For the remainder of the proof we will restrict ourselves to the time-homogeneous case; the time-inhomogeneous case follows by a straightforward extension of the arguments used in the time-homogeneous case and is left to the reader.

Before we proceed we introduce some further notation. The integral part of the infinitesimal generator, and its approximation will be denoted by

$$(A.33) \quad If(x) = \int_{(1,\infty)} [f(x(1+y)) - f(x) - xyf'(x)]g(x,y)dy,$$

$$(A.34) \quad I_n f(x_i) = \int_{(1,x_n/x_i-1)} [f_n(x_i(1+y)) - f_n(x_i)]g(x_i,y)dy - \mu_i x_i \nabla^- f(x_i),$$

where $\mu_i = \int_{(1,\infty)} yg(x_i,y)dy$, ' denotes the derivative with respect to x and

$$f_n(x) = \sum_{x_k \in (2x_i, \infty)} f(x_{k+1}) \mathbf{I}_{\left[\frac{x_k}{x_i}-1, \frac{x_{k+1}}{x_i}-1\right]}(x).$$

Furthermore, we define

$$(A.35) \quad I^{(k)} f(x) = \int_{(-1,1]} F^{(k)}(x,y)g(x,y)|y|^k dy,$$

$$(A.36) \quad I_n^{(k)} f(x_i) = \int_{(-1,1]} F_{n,i}^{(k)}(y)g(x_i,y)|y|^k dy,$$

where, for $k = 1, 2$,

$$\begin{aligned} F^{(k)}(x,y) &= |y|^{-k} [f(x(1+y)) - f(x) - f'(x)xy], \\ F_{n,i}^{(1)}(y) &= -x_i \tilde{\nabla} f(x_i) \text{sgn}(y) + \sum_{x_j \in [0, 2x_i] \setminus \{x_i\}} F_{n,ij}^{(1)} \mathbf{I}_{\left[\frac{x_j}{x_i}-1, \frac{x_{j+1}}{x_i}-1\right]}(y), \\ F_{n,i}^{(2)}(y) &= \sum_{(x_j/x_i-1) \in (-1,1) \setminus A_i^0(h)} \left[F_{n,ij}^{(2)} - \frac{x_i^2}{x_j - x_i} \nabla^0 f(x_i) \right] \mathbf{I}_{\left[\frac{x_j}{x_i}-1, \frac{x_{j+1}}{x_i}-1\right]}(y) \\ &\quad + \frac{x_i^2}{2} \Delta f(x_i) \mathbf{I}_{A_i^0(h)}(y), \end{aligned}$$

where $\tilde{\nabla}$ is defined in (A.24), with

$$F_{n,ij}^{(1)} = \frac{x_i}{|x_j - x_i|} \begin{cases} [f(x_{j+1}) - f(x_i)], & j > i, \\ [f(x_j) - f(x_i)], & j < i, \end{cases}$$

and

$$F_{n,ij}^{(2)} = \left(\frac{x_i}{x_j - x_i} \right)^2 \begin{cases} [f(x_j) - f(x_i)], & \frac{x_j}{x_i} - 1 \in (-1, 0) \setminus A_i^0(h), \\ [f(x_{j+1}) - f(x_i)], & \frac{x_j}{x_i} - 1 \in (0, 1) \setminus A_i^0(h). \end{cases}$$

Proof of Theorem 3. In view of Assumption 5, it follows that the stopped and discounted process $\tilde{S}^A = \{e^{-r(t \wedge \tau_A)} S_{t \wedge \tau_A}\}_{t \geq 0}$ is a Feller process. The infinitesimal generator of the stopped semi-group $\tilde{P}^A$ corresponding to $\tilde{S}^A$ is denoted by $\tilde{\mathcal{L}}^A$. The process $\tilde{S}^A$ is approximated by the Markov chains with infinitesimal generators $\tilde{\Lambda}_r^{(n)}$, where $\Lambda^{(n)}$ is described above and $\tilde{\Lambda}_r^{(n)}$ is obtained from $\Lambda^{(n)}$ by setting the rows of $\Lambda^{(n)}$ corresponding to A equal to zero and subtracting r on the diagonal elements of the rows corresponding to A^c , as in (3.6).

For any f satisfying Assumption 5, $\tilde{P}_t f$ has compact support as $\tilde{P}_t f(x) = f(x)$ for all $x \in A$ (recall we take $u < \infty$). Furthermore also by Assumptions 4 and 5 and the form of the Feller

process S under consideration $\tilde{P}_t f$ is contained in the domain $\mathcal{D}$ of the infinitesimal generator $\mathcal{L}$. Since

$$e^{-r(s \wedge \tau_A)} \tilde{P}_t f(S_{s \wedge \tau_A}) = E[e^{-r((s+t) \wedge \tau_A)} f(S_{(s+t) \wedge \tau_A}) | \mathcal{F}_s]$$

is a martingale, Dynkin's lemma implies that

$$(\mathcal{L} - r)(\tilde{P}_t f)(x) = 0, \quad x \in A^c,$$

so that in particular condition (2.13) holds true with f replaced by $\tilde{P}_t f$. From Lemma 1 and Assumption 5 we find that $\tilde{\mathcal{L}}^A(\tilde{P}_t f)$ is given by (2.12). Recall that, in the case under consideration, $\mathcal{L}f$ is of the form (2.11).

The form of the approximation and the triangle inequality imply that, for $H = \tilde{P}_s f$, $s \in [0, T]$, the distance $\epsilon'_n(H)$ between $\mathcal{L}$ and $\Lambda^{(n)}$ can be estimated as

$$\begin{aligned} \epsilon'_n(H) := \left\| \tilde{\Lambda}_r^{(n)} H - (\tilde{\mathcal{L}}^A - r)H \right\|_n &\leq \frac{1}{2} \|\Sigma^2\| \|\Delta_n H - H''\| + \|\Gamma\| \|\nabla_n H - H'\| + \|IH - I_n H\| \\ &+ \begin{cases} \|I^{(1)} H - I_n^{(1)} H\| & \text{if } \mathcal{I}_{i,j} < \infty, \\ \|I^{(2)} H - I_n^{(2)} H\| & \text{if } \mathcal{I}_{i,j} = \infty, \end{cases} \end{aligned}$$

where $\mathcal{I}_{i,j}$ is defined in (6.13). The norm $\|\cdot\|_n$ was defined in (A.31) and we wrote $\|\cdot\| = \sup_{x \in \hat{\mathbb{G}}(n)} |\cdot|$ for the supremum over all the points in the grid that lie between the barriers. The functions Σ and Γ are given by $\Sigma^2(x) = \sigma^2(x)x^2$ and $\Gamma(x) = \gamma x$ where $\gamma = r - d$. The operator Δ_n is equal to Δ defined in (A.17) and ∇_n is given by ∇^+ or ∇^- in (A.16) and (A.15) according to whether the constant γ is positive or negative, as in (A.14). We denote by H' the derivative of H with respect to x .

Writing

$$\|\cdot\|_\infty = \sup_{x \in \mathbb{E}} |\cdot|,$$

Assumptions 3, 4 and 5 and second and third order Taylor expansions yield that

$$\begin{aligned} \|\nabla_n H - H'\| &\leq \frac{h(n)}{2} \|H''\|_\infty \\ \|\Delta_n H - H''\| &\leq \frac{2h(n)}{3} \|H'''\|_\infty. \end{aligned}$$

Furthermore, the triangle inequality and Assumptions 3, 4 and 5 yield that, for $h(n)$ sufficiently small, we have

$$\begin{aligned}
\|IH - I_n H\| &\leq \max_{x_i \in \widehat{\mathbb{G}}^{(n)}} \left\{ \sum_{x_k \in (2x_i, x_n]} \int_{\left(\frac{x_k}{x_i} - 1, \frac{x_{k+1}}{x_i} - 1\right)} |H(x_i(1+y)) - H(x_{k+1})| g(x_i, y) dy \right. \\
&\quad \left. + \int_{\left(1, 1 + \frac{h(n)}{x_i}\right] \cup \left[\frac{x_n}{x_i} - 1, \infty\right)} H(x_n(1+y)) g(x_i, y) dy + |\mu_i| x_i |H'(x_i) - \nabla^- H(x_i)| \right\} \\
&\leq h(n) \cdot \|H'\|_\infty \cdot \max_{x_i \in \widehat{\mathbb{G}}^{(n)}} \int_{(1, \infty)} g(x_i, y) dy \\
&\quad + \|H\|_\infty \max_{x_i \in \widehat{\mathbb{G}}^{(n)}} \int_{\left(1, 1 + \frac{h(n)}{x_i}\right] \cup \left[\frac{x_n}{x_i} - 1, \infty\right)} g(x_i, y) dy + \frac{h(n)}{2} \|H''\|_\infty \max_{x_i \in \widehat{\mathbb{G}}^{(n)}} |\mu_i| x_i \\
&\leq C' \|H'\|_\infty \cdot h(n) + \|H\|_\infty \cdot \left(k(n) + \frac{h(n)}{\ell} \cdot \text{const} \right) + \frac{h(n)}{2} \|H''\|_\infty C' u,
\end{aligned}$$

where $\mu_i = \int_{(1, \infty)} y g(x_i, y) dy$, C' is given in Assumption 4, and $h(n)$ and $k(n)$ are the spatial mesh size and tail-mass defined in (6.17) and (6.18), and const is a bound for g on $[\ell, u] \times [1, 2]$. In case O the triangle inequality yields for n sufficiently large that

$$\|I^{(1)}H - I_n^{(1)}H\| \leq \tilde{C}_0 (\|H''\|_\infty + \|H'\|_\infty) h(n)$$

for some positive constant $\tilde{C}_0$. Indeed,

$$\begin{aligned}
&\text{(A.37)} \|I^{(1)}H - I_n^{(1)}H\| \\
&\leq \max_{x_i \in \widehat{\mathbb{G}}^{(n)}} \left\{ \sum_{x_k \in [0, 2x_i] \setminus \{x_i\}} \int_{\left(\frac{x_k}{x_i} - 1, \frac{x_{k+1}}{x_i} - 1\right)} \left| F^{(1)}(x_i, y) - F^{(1)}\left(x_i, \frac{x_k}{x_i} - 1\right) \right| g(x_i, y) y dy \right. \\
&\quad + x_i \left| \int_{(-1, 1]} y g(x_i, y) dy \right| \cdot |H'(x_i) - \tilde{\nabla} H(x_i)| \\
&\quad \left. + \int_{\left(0, \frac{x_{i+1}}{x_i} - 1\right)} \left| F^{(1)}(x_i, y) - x_i \tilde{\nabla} H(x_i) \right| g(x_i, y) y dy \right\} \\
&=: \max_{x_i \in \widehat{\mathbb{G}}^{(n)}} \{J_1(x_i) + J_2(x_i) + J_3(x_i)\}.
\end{aligned}$$

In view of Assumptions 3, 4 and 5 the three terms in (A.38) can be estimated as follows:

$$\begin{aligned}
J_1(x_i) &\leq \sum_{x_k \in (0, 2x_i], k \neq i} \int_{\left(\frac{x_k}{x_i} - 1, \frac{x_{k+1}}{x_i} - 1\right)} \left[\sup_{y \in \left(\frac{x_k}{x_i} - 1, \frac{x_{k+1}}{x_i} - 1\right)} \left| \frac{\partial F}{\partial y}(x_i, y) \right| \frac{x_{k+1} - x_k}{x_i} \right] g(x_i, y) y dy \\
&\leq \sum_{x_k \in (0, 2x_i], k \neq i} \int_{\left(\frac{x_k}{x_i} - 1, \frac{x_{k+1}}{x_i} - 1\right)} g(x_i, y) y dy \cdot \|H''\|_\infty \cdot x_i \cdot h(n) \\
&\leq \frac{3u}{2} \cdot C' \cdot \|H''\|_\infty \cdot h(n),
\end{aligned}$$

using that $|\frac{\partial F^{(1)}}{\partial y}(x, y)| \leq \frac{3x^2}{2} \|H''\|_\infty$, by a Taylor expansion. Furthermore, for $h(n)$ sufficiently small, we have

$$\begin{aligned} J_2(x_i) &\leq u \cdot \frac{h(n)}{2} \cdot \|H''\|_\infty \int_{(-1,1)} g(x_i, y) |y| dy \\ &\leq \frac{u}{2} \cdot \|H''\|_\infty C_0 \cdot h(n), \\ J_3(x_i) &\leq \frac{x_{i+1} - x_i}{x_i} \left(\frac{x_i^2}{2} \|H''\|_\infty + x_i \|H'\|_\infty \right) \int_{(0, \frac{x_{i+1}}{x_i} - 1)} yg(x_i, y) dy \\ &\leq h(n) \cdot \left(\frac{u}{2} \|H''\|_\infty + \|H'\|_\infty \right) \cdot C_0 \end{aligned}$$

where we used that for $y \in (x_i, x_{i+1})$, $|F^{(1)}(x_i, y)| \leq \frac{x_i^2}{2} \|H''\|_\infty$, by a Taylor expansion.

In cases I and II,

$$\begin{aligned} &\|I^{(2)}H - I_n^{(2)}H\| \\ &\leq \max_{x_i \in \widehat{\mathbb{G}}^{(n)}} \left\{ \sum_{x_k \in (-1,1) \setminus A_i^0(h)} \int_{(\frac{x_k}{x_i} - 1, \frac{x_{k+1}}{x_i} - 1)} \left| F^{(2)}(x_i, y) - F^{(2)}(x_i, \frac{x_k}{x_i} - 1) \right| g(x_i, y) y^2 dy \right. \\ &\quad + \sum_{x_k \in (-1,1) \setminus A_i^0(h)} \frac{x_i^2}{|x_k - x_i|} |H'(x_i) - \nabla^0 H(x_i)| \cdot \int_{(\frac{x_k}{x_i} - 1, \frac{x_{k+1}}{x_i} - 1)} y^2 g(x_i, y) dy \\ &\quad \left. + \int_{A_i^0(h)} \left| F^{(2)}(x_i, y) - \frac{x_i^2}{2} \Delta H(x_i) \right| g(x_i, y) y^2 dy \right\} \\ (A.38) &=: \max_{x_i \in \widehat{\mathbb{G}}^{(n)}} \{K_1(x_i) + K_2(x_i) + K_3(x_i)\}. \end{aligned}$$

In view of Assumptions 3, 4 and 5 the three terms in (A.38) can be estimated as follows:

$$\begin{aligned} &K_1(x_i) \\ &\leq \sum_{(x_k/x_i - 1) \in (-1,1) \setminus A_i^0(h)} \frac{x_{k+1} - x_k}{x_i} \cdot \max_{y \in (\frac{x_k}{x_i} - 1, \frac{x_{k+1}}{x_i} - 1)} \left| \frac{\partial F^{(2)}}{\partial y}(x_i, y) \right| \cdot \int_{(\frac{x_k}{x_i} - 1, \frac{x_{k+1}}{x_i} - 1)} g(x_i, y) y^2 dy \\ &\leq \frac{5}{6\ell} \cdot C'' \cdot \|H'''\|_\infty \cdot h(n), \end{aligned}$$

where C'' is a constant that satisfies

$$(A.39) \quad C'' \leq 3C'$$

for h sufficiently small and the bound

$$\frac{\partial F^{(2)}}{\partial y}(x, y) = \frac{\partial}{\partial y} \frac{H(x(1+y)) - H(x) - H'(x)xy}{y^2} \leq \frac{5x^3}{6} \cdot \max_{z \in [1 \wedge (1+y), 1 \vee (1+y)]} |H'''(xz)|$$

holds. Furthermore we find

$$\begin{aligned} K_2(x_i) &\leq x_i^2 \cdot \|H'''\|_\infty \cdot \sum_{(x_k/x_i - 1) \in (-1,1) \setminus A_i^0(h)} \left| \frac{(x_{i+1} - x_i)(x_i - x_{i-1})}{x_k - x_i} \right| \int_{(\frac{x_k}{x_i} - 1, \frac{x_{k+1}}{x_i} - 1)} g(x_i, y) y^2 dy \\ &\leq u^2 \|H'''\|_\infty \cdot C'' \cdot h(n). \end{aligned}$$

Note that the equality

$$F^{(2)}(x_i, y) - \frac{x_i^2}{2} \Delta H(x_i) = \frac{x_i^2}{2} [H''(x_i) - \Delta H(x_i)] + \frac{x_i^3 y}{6} H'''(\xi_y x_i),$$

for some $\xi_y \in [1 \wedge (1 + y), 1 \vee (1 + y)]$, implies that

$$K_3(x_i) \leq \frac{x_i^3}{3!} \cdot \|H'''\|_\infty \cdot \int_{A_i^0(h)} |y|^3 g(x_i, y) dy \leq \frac{u^2}{3!} \cdot C' \cdot \|H'''\|_\infty \cdot h(n)(C_+(h(n)) + C_-(h(n)) + 2).$$

Putting everything together we find that

$$(A.40) \quad \begin{aligned} \epsilon'_n(H) &\leq h(n)A_0 [\|H'''\|_\infty + \|H''\|_\infty + \|H'\|_\infty + \|H\|_\infty] + 2k(n)\|H\|_\infty \\ &+ h(n) \begin{cases} A_1, & \text{cases 0 and II,} \\ A_2 + A_3 |\log(h(n))|, & \text{case I} \end{cases} \end{aligned}$$

for some positive constants A_i , $i = 0, \dots, 3$. All supremum norms in (A.40) are finite, since they are equal to the maxima of continuous functions over compact sets. Combining (A.40) with Lemma 3, which is applicable in view of the assumptions, the theorem follows in the time-homogeneous case. $\square$

APPENDIX B. MATLAB CODE

This appendix contains the Matlab code for the pricing of barrier options under the CEV model. The program consists of the following three files, which can also be downloaded from [51]:

```
main.m
generate_diffusion_MG.m
generate_grid_for_double_barrier.m
```

The content of these files can be found below. The function `main` takes no parameters and prints out the prices of the 3 double barrier options in the CEV model that correspond to 3 different strikes and have the same barrier levels. The parameter values can be changed in the file `main.m`.

The function `generate_diffusion_MG` computes a generator matrix for a given state-space. The file `generate_grid_for_double_barrier.m` generates a non-uniform grid for the approximating chain. The driving routine is `main.m`; to run the programme type

```
>> main
```

in the Matlab prompt.


```

100, 1, ...
10, 10, ...
1, 100);

%   compute the markov generator MG

MG = generate_diffusion_MG(s_grid, spotS, vol, rate, beta);

%   set up the call payoff

for y=1:size(strikes,2)
    payoff(:,y) = max(s_grid - strikes(y),0)';
end

%   compute the barrier generator and payoff

barrier_idx = find(L<s_grid & s_grid<U);
BarrierMG = MG(barrier_idx,barrier_idx);
bar_s_grid = s_grid(barrier_idx);
newspotS_index = find(bar_s_grid == spotS);

%   compute the barrier option prices

A = expm(T*BarrierMG); % exponentiate the barrier generator
barP = A(newspotS_index,:);
barPayoff = payoff(barrier_idx,:);
BarrierPrices = exp(-rate * T) * barP * barPayoff;

toc

fprintf('\n')
fprintf('----- CEV - Double Barrier Call Options -----\n')
fprintf('time to maturity:')
disp(T);
fprintf('strikes:')
disp(strikes);

```


[illegible]

Electronic copy available at: <https://ssrn.com/abstract=1462822>

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